CSE 450 Capstone Project Report

Leveraging Temporal Information for Robust Liveness Verification

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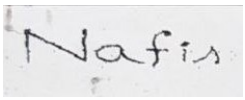
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Candidates' Declaration

We declare that this report presents our own capstone work except where sources, datasets, pretrained models, software, or ideas are explicitly acknowledged. To the best of our knowledge, it has not been submitted in the same form for another degree or award. Each candidate accepts responsibility for the integrity of the reported methods, results, and limitations.

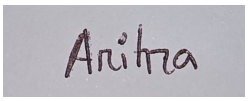
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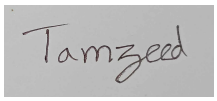
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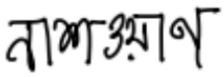
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We thank Pathao for presenting the original financial liveness-verification problem and for the early project context. The subsequent research and implementation were carried out independently using public datasets and our own computing resources; no private Pathao dataset, project grant, or sustained technical collaboration was received.

We also acknowledge the authors and maintainers of the public datasets, research implementations, and open-source libraries on which the experiments depended. Finally, we thank our teachers, classmates, friends, and families for their feedback and support throughout the capstone.

Abstract

Remote financial identity verification must distinguish genuine users from both physical presentation attacks and digitally manipulated or injected face media. This project investigates a layered liveness-verification architecture whose first stage is a lightweight temporal MobileNetV3 model for mobile inference and whose stronger experimental components include G2V2Former temporal reasoning, GD-FAS domain-generalised spatial features, Xception deepfake detectors, and a learned manipulation-specific routing strategy. The mobile prototype captures a short clip, samples 24 frames, performs local inference, and applies a fixed decision threshold. Server escalation, score fusion, capture attestation, and human review are target-architecture elements rather than completed integrations.

Cross-domain experiments reveal the central difficulty. Under the repository's common evaluation protocol, GD-FAS trained on LCC-FASD achieved 4.55% HTER and 99.69% AUC on LCC-FASD but 52.99% HTER and 50.49% AUC on Celeb-DF-v2. Fine-tuning on Celeb-DF-v2 improved its target-domain result to 41.04% HTER and 64.35% AUC while reducing source-domain performance. G2V2Former showed the same generalisation challenge. A later FaceForensics++ experiment compared a universal Xception detector with manipulation-specific experts selected by a learned, best-expert router that estimates the most suitable one of five binary specialists from an embedding, with no "original" routing class. The routed system improved balanced accuracy from 76.44% to 84.51% and ROC-AUC from 86.32% to 91.68% on a group-disjoint, in-domain validation split.

The work contributes an evidence-based analysis of temporal information, cross-domain transfer, mobile deployment constraints, and specialist routing. It also identifies the safeguards required before financial use: capture-channel integrity, subgroup and cross-domain evaluation, privacy protection, calibrated thresholds, accessible appeal, and accountable human oversight.

Keywords: face liveness; deepfake detection; temporal modelling; domain generalisation; specialist routing; mobile inference.

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Chapter 1

Problem Identification and Formulation

1.1 Background and Motivation

Remote onboarding and account recovery frequently compare a selfie or short face video with identity evidence. The comparison is useful only if the media came from the person and capture session being verified. An attacker may instead present a printed photograph, replay a video on another screen, wear a mask, display a manipulated video, or bypass the physical camera and inject forged media. ENISA consequently separates camera presentation from injection attacks and recommends layered controls spanning the client and server [1, 2]. NIST likewise treats sensor authenticity, protected channels, presentation-attack detection (PAD), and forged-media analysis as complementary controls rather than substitutes [3].

This problem is especially consequential in financial verification. A false acceptance can enable impersonation and monetary loss, while a false rejection can exclude a legitimate customer or repeatedly expose their biometric data. The system must therefore balance security, usability, latency, resource use, privacy, and equitable performance instead of optimising only dataset accuracy.

Deepfake attacks are difficult for several connected reasons:

- manipulation mechanisms are heterogeneous and continue to evolve;
- compression, resizing, recapture, and post-processing can erase the artefacts on which a detector relies;
- a model can learn dataset, identity, background, or codec shortcuts and then fail under a new generator or population;
- a manipulated face can preserve plausible blinking, pose, and motion, so conventional liveness cues alone do not establish authenticity;
- a virtual camera or compromised client can bypass assumptions about a trusted physical capture path; and
- mobile inference restricts model size, memory, energy, and response time.

This is not a hypothetical threat model: industry threat-intelligence monitoring of face-biometric KYC systems has recorded face-swap and digital-injection attacks growing as separate, independently accelerating categories rather than variants of the same print/replay problem [4], which is exactly why Figure 1.1 keeps delivery path and manipulation family on separate axes instead of one severity scale, and why capture-channel assurance (Section 4.5) is tracked as a requirement distinct from classification accuracy: no amount of accuracy on a presented frame defends against media that never passed through the physical camera at all. It is specifically the growth in face-swap and reenactment attacks—rather than print or replay—that motivates giving the system a dedicated digital-manipulation layer at all: an on-device presentation-attack screen (Layer 1) has no mechanism to see this class of attack, since a face swap presents a geometrically and temporally plausible live capture. Layer 2 (Section 4.1) closes exactly this gap by routing every frame that Layer 1 accepts to a manipulation-specific Xception specialist; the per-method results in Table 5.4 show that this routed check raises fake-class detection on the hardest evaluated method (NeuralTextures) from a near-chance 72.66% to 76.11% balanced accuracy under a universal detector, and above 85% for the four identity-swap and reenactment methods closest to a real face-swap attack. Because Table 4.2 requires both layers to agree, a face swap that fools the on-device screen must still pass Layer 2's specialist check to be accepted, which is the strict, conjunctive sense in which Layer 2 constrains the system's exposure to this growing attack class—strict within the scope this report can actually claim, since Section 5.2 still limits that claim to the in-domain, group-disjoint FF++ evaluation and not to unseen generators or the injection attacks discussed above.

The attack must consequently be described along more than one axis. Figure 1.1 distinguishes the delivery path from the manipulation family. The five FaceForensics++-derived experimental fake classes are not five equivalent meanings of "deepfake": Deepfakes, FaceSwap, and FaceShifter are identity-swap approaches, whereas Face2Face and NeuralTextures are reenactment/neural-rendering approaches [5, 6]. "Original" denotes genuine data and is not a manipulation class.

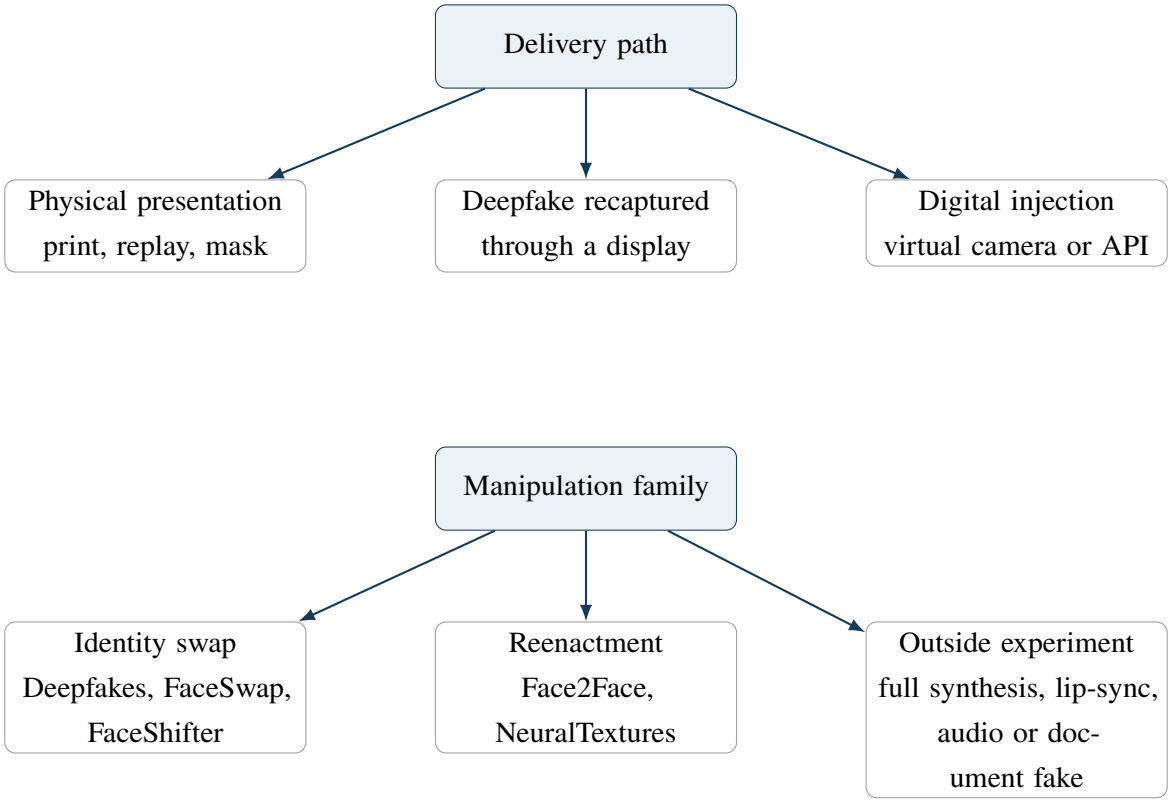
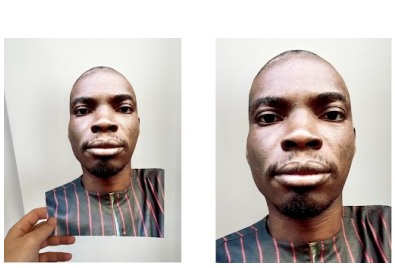
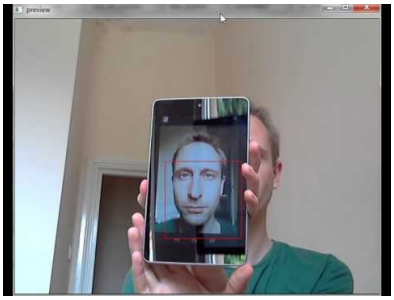


Figure 1.1: Threat taxonomy used in this report. Delivery path and manipulation family are separate: the same generated video may be shown to a camera or injected directly. Items outside the experiment remain relevant threats.

Figure 1.2 shows one captured or sourced example of each of the three delivery paths actually exercised while building and testing the prototype: a printed photograph held in front of the camera, a video replayed on a second screen, and a face-swapped frame paired with its genuine source. All three defeat a plain single-frame camera check, which is why the taxonomy in Figure 1.1 treats delivery path and manipulation family as separate axes rather than a single attack label.



(a) Print attack



(b) Replay attack



(c) Deepfake (identity swap)

Figure 1.2: Sample attacks along each delivery path. (a) A printed photograph held up to the camera. (b) A video replayed on a second screen, captured by the verifying device. (c) A genuine frame (left) and a face-swapped frame (right) of the same clip; the swap is convincing in a single frame.

1.2 Problem Statement

The engineering problem is to design and evaluate a practical face-liveness screening pipeline for remote financial verification that detects both physical presentation attacks and digitally manipulated faces, uses temporal evidence where useful, degrades safely under uncertainty, and remains feasible on a mobile–server architecture. The work asks four linked questions:

1. Can a lightweight temporal model provide a useful first decision on a mobile device?
2. How well do stronger temporal and domain-generalised PAD models transfer from physical-spoof datasets to deepfake datasets?
3. Does manipulation-specific expertise improve frame-level deepfake detection relative to one universal detector?
4. What security and ethical controls are required before experimental accuracy can become a trustworthy financial decision?

For a captured clip $V = (x_1, \ldots, x_K)$, the first-layer model produces a fake probability $p_1 = p(\text{fake} \mid V)$. A target cascade uses two policy thresholds $\tau_{\text{pass}} < \tau_{\text{attack}}$: clear low-risk cases may pass, clear high-risk cases may be rejected or challenged, and the interval between them is escalated. These thresholds are a design requirement; the current Flutter prototype applies a single threshold of 0.5.

For the later specialist experiment, let x be a preprocessed face frame, $z = \phi(x)$ an embedding, and $\mathcal{E}$ the five manipulation-specific binary experts. The router is

$$q_e(z) = P(E^* = e \mid z), \qquad e \in \mathcal{E}, \tag{1.1}$$

$$\hat{e} = \arg\max_{e \in \mathcal{E}} q_e(z), \tag{1.2}$$

$$\widehat{p}(\text{fake} \mid x) = s_{\hat{e}}(x), \tag{1.3}$$

where s_e is the fake probability from expert e. The router does not predict "original"; every frame is assigned to a binary specialist. The training target E^* for each frame is the best-expert label: every frame is scored by all five already-trained experts, and the expert with the highest score against the frame's true class becomes that frame's router label. This is a best-expert formulation rather than a manipulation-method label, so the learned quantity is $P(\text{best expert} \mid z)$, not $P(\text{manipulation method} \mid z)$.

Formulated in engineering terms:

Table 1.1: The problem stated in engineering terms.

Element	Specification
Inputs	A short front-camera facial clip captured on a mobile device; a single preprocessed face frame for the manipulation-specialist experiment; deployment-time policy thresholds $\tau_{\text{pass}}, \tau_{\text{attack}}$ (target-state)
Outputs	A fake probability and a thresholded real/fake decision for the mobile clip; a routed fake probability per frame for the specialist experiment; target-state: a calibrated PASS/FAIL/REVIEW decision with a reason code
Entities	Applicant, capture session, video frame, facial landmark, manipulation method, binary specialist, meta-router, model checkpoint, reviewer (proposed)
Quantities guaranteed	None formally guaranteed by the current prototype; local mobile inference completes without a network dependency; validation for the FF++ routing experiment is disjoint at the source-video group level
Quantities optimised	Balanced accuracy, ROC-AUC, average precision, and class-conditional real/fake accuracy for the routing experiment; HTER, APCER, and BPCER for the physical-versus-digital transfer study; mobile latency and model footprint
Assumptions	Genuine capture originates from the device's physical camera, since no capture-channel assurance is implemented; public benchmark datasets approximate the deployment population and attack distribution; one frame carries enough evidence for the manipulation-specialist routing decision

1.3 Complexity of the Problem

A complex engineering problem must have characteristic P1 and some or all of P2–P7. P1 requires in-depth knowledge at the level of one or more of the knowledge profiles K3–K6 and K8 listed in Table 1.3. Table 1.2 records project-specific evidence for each characteristic.

Table 1.2: Complex engineering problem characteristics (P1–P7) and evidence from this project.

Code	Definition	Evidence from this project
P1	**Depth of knowledge required.** Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8.	**K3, K4, K5, K6, K8.** K3: probability and statistics underlying the HTER/APCER/BPCER and ROC-AUC metrics used throughout Chapter 5. K4: deep-learning specialist knowledge spanning convolutional backbones (Xception, MobileNetV3), graph-guided temporal attention (G^2V^2former), and domain-generalisation theory (GD-FAS), Section 2.1. K5: knowledge supporting the design of the layered mobile–server cascade, Section 4.1. K6: current practice in mobile-model export and biometric-security guidance (ISO/IEC 30107-3, NIST SP 800-63A-4), Chapter 3. K8: engagement with the FaceForensics++, Celeb-DF-v2, G^2V^2former, and GD-FAS research literature, Section 2.1.
P2	**Range of conflicting requirements.** Involves wide-ranging or conflicting technical, engineering, and other issues.	Security against both attack families versus low user friction; mobile latency and memory versus model capacity; per-manipulation specialisation versus generalisation to unseen generators; on-device privacy versus stronger server-side analysis. Each is addressed explicitly in Section 3.3 and Chapter 4.
P3	**Depth of analysis required.** Has no obvious solution and requires abstract thinking and originality in analysis to formulate suitable models.	No single detector transfers across physical presentation attacks and digital manipulation attacks (Table 5.1); interpreting why source-domain-strong models collapse under this shift, and formulating the embedding-conditioned routing hypothesis, required original analysis (Section 4.2.3).
P4	**Familiarity of issues.** Involves infrequently encountered issues.	Unseen-generator generalisation, a capture channel bypassing the physical sensor via injection, numerical instability in attention-guided temporal training (the G^2V^2former fix, Section 4.2), and best-expert labelling for the router (Section 4.2.3) are all non-routine.
P5	**Extent of applicable codes.** Is outside problems encompassed by standards and codes of practice for professional engineering.	ISO/IEC 30107-3 defines PAD testing but explicitly excludes injection outside the capture device (Section 3.2); NIST SP 800-63A-4 and ENISA guidance address part of that gap but are not binding Bangladeshi law. The project therefore defines its own Implemented/Experimental/Proposed status labels where no code fully specifies digital-deepfake or injection defence.

Table 1.3: Knowledge profiles referenced by P1.

Code	Description
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Knowledge that supports engineering design in a practice area.
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K8	Engagement with selected knowledge in the research literature of the discipline.

1.4 Project Objectives and Scope

The project set the following objectives. Table 5.5 evaluates the delivered work against each.

O1. Establish a clear physical-presentation, digital-manipulation, and injection threat model for remote face verification.

O2. Build a lightweight temporal first-stage classifier and integrate its exported model into a Flutter capture flow.

O3. Investigate richer temporal evidence through G^2V^2former.

O4. Compare temporal modelling with the GD-FAS domain-generalisation method.

O5. Measure within-domain and cross-domain behaviour on public PAD and deepfake datasets.

O6. Compare a universal Xception detector with manipulation-specific specialists and learned routing.

O7. Specify a layered escalation architecture and the controls needed for trustworthy deployment.

O8. Analyse privacy, fairness, accessibility, accountability, intellectual property, cost, and environmental impact.

The implemented scope comprises public-data experiments, preprocessing and training notebooks, a Flutter/mobile temporal prototype, a separately demonstrated G^2V^2former API, GD-FAS evaluation, and frame-level Xception routing experiments. It does not comprise a production Pathao Pay deployment, a connected multi-model backend, capture-device attestation, a reviewer console, calibrated production thresholds, or a completed field trial. Audio deepfakes, document

forgery, fully synthetic identities, and end-to-end identity matching are threat context but not separately evaluated model classes.

Chapter 2

Preliminary Research

2.1 Literature Review

2.1.1 Presentation attacks, deepfakes, and injection

PAD traditionally evaluates whether a biometric sample is bona fide or a presentation attack at the capture device. ISO/IEC 30107-3 defines principles for testing and reporting PAD performance, including attack-type classification, but explicitly excludes attacks outside the capture device from its scope [7]. This boundary matters: a detector that recognises screen moiré or print texture does not establish that a video stream came from a genuine camera. ENISA's later guidance therefore treats injection attack detection (IAD) alongside PAD and recommends a multi-layer client–server approach [2]. NIST SP 800-63A-4 similarly calls for confidence in the genuine sensor, protected communications, manipulated-media analysis, and documented manual processes [3].

FaceForensics++ provides a controlled benchmark with several facial manipulation methods and compression levels, and established Xception as a strong baseline [5, 8]. Celeb-DF-v2 was designed with higher-quality face swaps and fewer conspicuous synthesis artefacts, making it a more challenging test of shortcut-dependent detectors [9]. Cross-dataset deepfake work repeatedly shows that strong in-domain performance does not guarantee unseen-generator performance; video-oriented approaches such as AltFreezing were developed specifically to improve this transfer [10].

2.1.2 Temporal and domain-generalised face anti-spoofing

MobileNetV3 uses hardware-aware architecture search and efficient blocks intended for resource-constrained inference [11]. The project's on-device stage adapts STDL-FacePAD, which applies MobileNetV3 independently to sampled frames and averages the resulting features by adaptive temporal pooling before classification [12]. STDL-FacePAD's own evaluation, on Replay-Attack, Replay-Mobile, and ROSE-Youtu, reports equal-error rates of 0.0%, 0.0%, and 0.83% respectively with an 11 ms inference time for a 256-sample batch; those figures describe physical presentation-attack datasets similar in character to LCC-FASD and should not be read as deepfake-detection

performance. Averaging is efficient and permutation-invariant, but it cannot represent temporal order, which motivates the richer temporal stage below.

G^2V^2former combines cropped face appearance with facial landmarks in a two-stream encoder (Figure 2.1). Each stream stacks spatiotemporal blocks of a spatial-attention module followed by a temporal-attention module; the vision stream's spatial attention is guided by a photometric-consistency loss, while its temporal attention uses a Kronecker temporal attention that the graph stream's landmark motion helps steer through a scatter-and-add guidance signal. Both streams are pretrained independently with masked reconstruction before joint supervised fine-tuning [13]. It is therefore relevant when motion contains evidence that is weak or ambiguous in any individual frame.

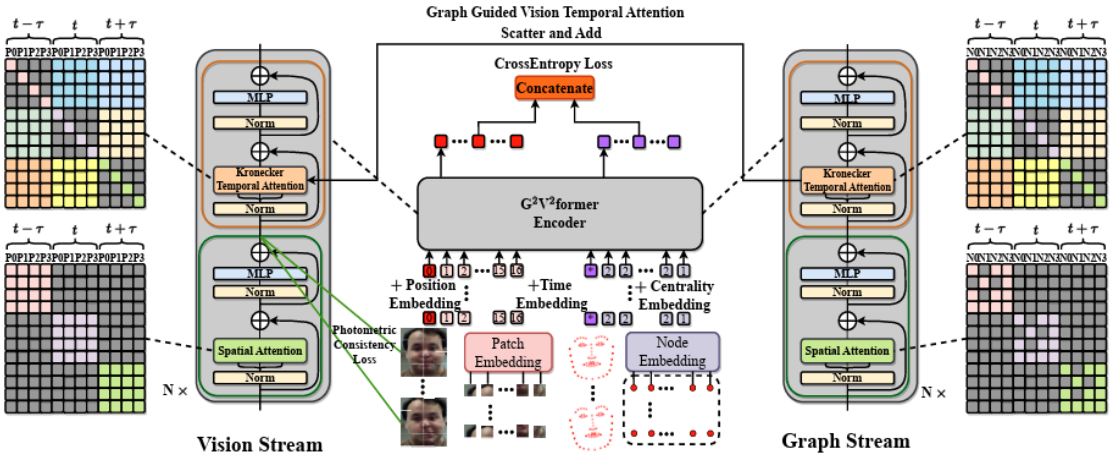


Figure 2.1: G^2V^2former architecture: vision and graph streams of stacked spatiotemporal blocks, fused through graph-guided temporal attention and a concatenated classification head. Reproduced from [13].

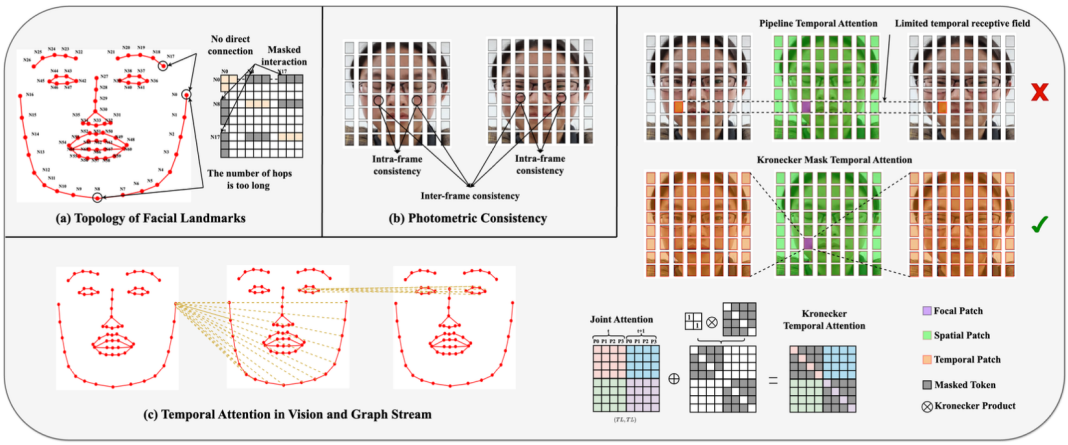


Figure 2.2: Why the vision stream needs graph guidance rather than plain temporal attention: the facial-landmark topology limits which nodes may interact (a), photometric consistency defines the vision-stream loss (b), and Kronecker temporal attention is needed because ordinary pipeline temporal attention has a limited temporal receptive field (c). Reproduced from [13].

This mechanism is built specifically around physical presentation-attack evidence—landmark motion, photometric consistency across frames, patch-level temporal correlation—and, as Section 5.2.1 shows quantitatively, that specificity is also the architecture's limitation: it is biased toward the attack family it was designed and trained against and does not, on its own, transfer to digitally generated manipulations.

GD-FAS addresses a different failure mode (Figure 2.3). Built on a CLIP image/text encoder pair, its group-wise scaling risk minimisation and feature orthogonal decomposition align both the direction and bias of local decision boundaries while separating domain-invariant from domain-specific features [14]. It remains a frame-level spatial model in this project; comparing it with G^2V^2former helps separate the value of temporal structure from the value of domain-generalised representation.

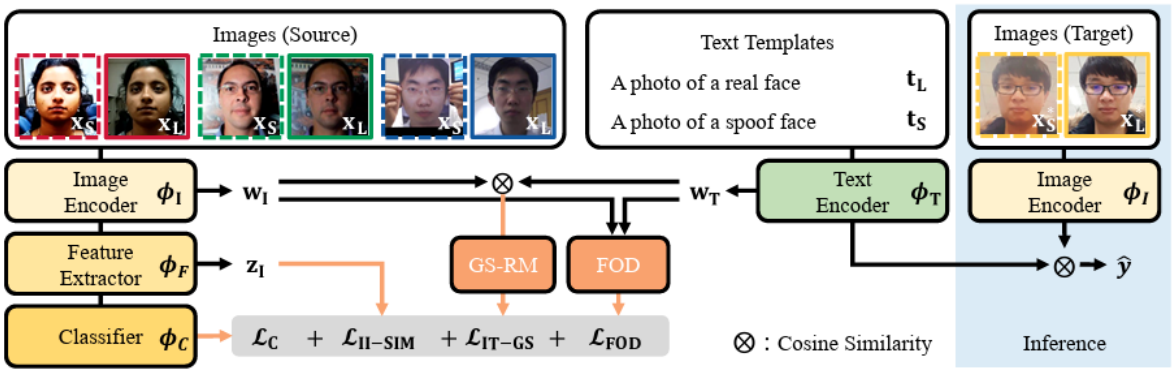


Figure 2.3: GD-FAS framework: CLIP image and text encoders feed a feature extractor and classifier regularised by group-wise scaling (GS-RM) and feature orthogonal decomposition (FOD). Reproduced from [14].

Table 2.1: Research approaches and their roles in the project.

Approach	Primary evidence	Strength	Limitation relevant here
MobileNetV3 pooling	Short face clip	Efficient mobile temporal summary	Mean pooling can discard ordering and subtle motion.
G^2V^2former	Face video and 68 landmarks	Explicit photometric–dynamic fusion	Landmark extraction and attention increase latency and failure modes.
GD-FAS	Single face frames and domain labels	Explicit domain-invariant feature objective	Does not directly model temporal consistency.
Universal Xception	Single face frame	Simple common detector and baseline	A single boundary may average incompatible manipulation artefacts.
Routed specialists	Embedding plus expert bank	Allocates capacity by manipulation pattern	Router errors, training leakage, and compute multiply risk.
Capture/IAD controls	Device and channel signals	Addresses camera bypass	Not implemented by a media classifier alone.

2.2 Existing Solutions and Technologies

Commercial liveness systems commonly advertise passive analysis, active challenge–response, or combinations of both. Active systems ask the user to perform a random action; they may provide session-specific evidence but add friction and accessibility concerns. Passive systems analyse ordinary capture in the background; they can improve usability but must not be treated as proof of a trusted source. Vendor claims are useful indicators of product direction, not independent evidence of accuracy.

The technological alternatives considered were lightweight convolutional networks, video transformers, domain-generalisation methods, universal deepfake detectors, specialist ensembles, active challenges, and capture-channel controls. No one option covers every threat. The selected design therefore treats the classifier as one layer in a risk-based verification process.

2.3 Market and User Research

Pathao supplied the initial financial-verification problem and early context, but the team did not receive private operational data, a grant, or sustained product collaboration. Consequently, this report does not claim direct Pathao user research, production integration, or measured business impact. The requirements instead derive from the initial problem framing, security guidance, public literature, and prototype observation.

The affected user groups extend beyond fraud analysts. Legitimate applicants need a quick, comprehensible process that works across devices, connectivity levels, skin tones, ages, abilities, and religious or cultural presentation. Reviewers need bounded workloads and enough context to avoid blind trust in scores. A financial provider needs auditable decisions, recoverable failure handling, and controls against both fraud and unlawful or excessive biometric processing.

Before deployment, the market/user study must therefore add interviews or tests with applicants, accessibility participants, fraud reviewers, support personnel, security engineers, privacy/legal personnel, and product owners. Required outputs include acceptable capture time, retry limits, low-bandwidth behaviour, consent language, review turnaround, and an explicit false-accept/false-reject cost model.

2.4 Feasibility Study

Technical feasibility. A temporal MobileNetV3 has been exported and integrated into a Flutter capture path, showing that local inference is feasible. The stronger models run in research/server environments. End-to-end selective escalation and capture attestation remain unimplemented, so system feasibility is only partially demonstrated.

Data feasibility. Public datasets make repeatable research possible, but they cannot represent the final operating population or attack distribution. Licences, consent conditions, identity leakage, demographic composition, and subject-disjoint splits must be audited before reuse. No private Pathao data is available for deployment-domain validation.

Operational feasibility. A cascade can reduce server load by retaining a fast local path, but only after scores are calibrated and thresholds are selected from operational costs. Network failure, poor capture, device compromise, and review capacity require explicit fallback behaviour.

Economic and legal feasibility. Open-source tooling and public data reduce prototype cost, whereas GPU training, secure infrastructure, testing, monitoring, and human review remain material. NIST and ENISA are authoritative engineering guidance, not declarations of legal compliance in Bangladesh. Local privacy, financial, cyber-security, consumer-protection, and evidentiary requirements require review by qualified stakeholders before deployment.

Conclusion. Research prototyping is feasible, and the on-device stage has been demonstrated. Production adoption is conditional on integrated security controls, field data, independent PAD/IAD testing, subgroup analysis, operational calibration, and governance.

Chapter 3

Requirements Analysis

3.1 Project Objectives and Constraints

The requirements are expressed at system level so that a model metric is not mistaken for a complete security property. Table 3.1 records the evidence currently available. “Partial” means that an artefact exists but the full requirement has not been demonstrated.

Table 3.1: Requirement and evidence matrix.

ID	Requirement	Status	Current evidence
FR-01	Capture a short facial video and perform basic capture validation before making a verification decision.	**Implemented**	The Flutter application captures the facial clip, while ML Kit face tracking and active liveness actions gate the capture before model inference.
FR-02	Run a lightweight temporal first-stage model locally without requiring the complete verification pipeline to execute on the server.	**Implemented**	The exported MobileNetV3 PyTorch Lite model runs locally in Flutter using a 24-frame input sequence and produces the first-stage liveness decision.
FR-03	Produce a model confidence score and an explicit real/spoof decision for the captured session.	**Implemented**	The on-device model produces a probability score and applies the configured 0.5 threshold to generate the local real/spoof verdict.

ID	Requirement	Status	Current evidence
FR-04	Send captured evidence to a stronger server-side analysis stage in parallel with the lightweight local detector.	**Implemented**	A captured frame is sent to the FastAPI backend while MobileNetV3 runs locally; the server executes the router and selected Xception specialist and returns its deepfake verdict.
FR-05	Detect both physical presentation attacks and digitally manipulated face media within the verification pipeline.	**Implemented**	Physical presentation attacks are handled by the on-device liveness/spoof stage, while digitally manipulated faces are analysed by the server-side Xception deepfake-detection stage.
FR-06	Detect camera bypass or digitally injected media that does not originate from the genuine physical camera.	**Proposed**	Injection attacks are included in the threat model, but genuine-camera attestation and dedicated injection-attack detection are not implemented.
FR-07	Route server-side face evidence to a manipulation-specific Xception specialist using a learned routing model.	**Implemented**	The deployed server uses the trained five-way router to identify the routing class and invokes the corresponding manipulation-specific Xception specialist for the final deepfake score.
FR-08	Handle unsuccessful verification safely by producing a flagged/rejected state and allowing the user to retry.	**Implemented**	If either the local spoof detector or server-side deepfake detector rejects the sample, the client produces a flagged/rejected result and directs the user through the retry flow rather than silently accepting the session.

ID	Requirement	Status	Current evidence
NFR-01	Maintain video-group separation between training and validation data in the FaceForensics++ routing experiments to reduce data leakage.	**Implemented**	The FaceForensics++ experiment separates training and validation at the source-video-group level rather than randomly splitting individual frames.
NFR-02	Evaluate the developed models using metrics appropriate to liveness and deepfake detection rather than relying only on raw accuracy.	**Implemented**	The experiments report measures including HTER, AUC, balanced accuracy, average precision, and class-conditional performance as appropriate to the individual evaluation protocols.
NFR-03	Reduce unnecessary biometric-data transfer by performing the lightweight first-stage liveness decision locally on the mobile device.	**Implemented**	MobileNetV3 inference is performed on-device from the captured 24-frame sequence; only the evidence required for the server-side deepfake stage is sent to the backend.
NFR-04	Bound and evaluate latency, memory, battery, bandwidth, and server-side computational cost.	Partial	Latency was measured for the mobile pipeline and several server execution configurations, but full battery, memory, bandwidth, and long-term resource profiling has not been completed.
NFR-05	Expose sufficient runtime decision information for debugging, evaluation, and tracing of the verification pipeline.	**Implemented**	The server response exposes the deepfake probability, routed class, and per-call latency, while the application retains the local and server verdicts needed to produce the final decision.

ID	Requirement	Status	Current evidence
NFR-06	Provide a recoverable user flow when verification fails instead of producing an ambiguous or silent failure.	**Implemented**	Failed local or server verification produces an explicit rejected/flagged state in the Flutter application and provides a retry path for the user.

Key constraints are the lack of private deployment data, limited GPU access, heterogeneous dataset licences and annotations, mobile compute limits, and the changing attack landscape. The most important safety constraint is epistemic: a score produced on a public benchmark cannot be represented as fraud probability for a real applicant without deployment-domain validation and calibration.

3.2 Regulatory Requirements and Standards

Table 3.2: Standards and guidance relevant to the target system.

Source	Relevance to the project	Application in this work
ISO/IEC 30107-3:2023 [7]	Provides principles for presentation-attack detection (PAD) testing and reporting, including the treatment of different attack-presentation types.	Used as a reference for PAD evaluation and terminology. No certification or formal conformance is claimed, and attacks occurring outside the biometric capture device, such as direct digital injection, are outside its PAD scope.
NIST SP 800-63A-4 [3]	Provides guidance for remote identity proofing, including presentation-attack detection, confidence in the capture sensor, protected communication channels, injection attacks, and manipulated-media detection.	Used to guide the security requirements and layered design of the proposed verification system. It is treated as technical guidance rather than as binding Bangladeshi law.
ENISA remote-ID reports [1, 2]	Describe threats to remote identity verification, including presentation and injection attacks, and recommend layered client–server countermeasures.	Used to develop the project threat model and motivate separation between physical presentation attacks, digital manipulation, and capture-channel attacks. The reports are used as security guidance rather than as product certification.
ACM/IEEE ethical codes [15, 16]	Emphasise avoidance of harm and discrimination, protection of privacy, honesty about system limitations, professional competence, and accountability.	Used as the basis for the project's discussion of privacy, fairness, responsible reporting, human oversight, and professional responsibility.
Applicable Bangladesh legal and financial requirements	Relevant to biometric-data processing, cyber security, consumer protection, financial records, user consent, and mechanisms for review or appeal.	A production deployment would require review by appropriate legal, financial, privacy, and compliance stakeholders. This report does not claim legal compliance or provide a legal interpretation of Bangladeshi law.

For a production deployment, regulatory and security requirements extend beyond the accuracy of the liveness models. The system should support purpose limitation, data minimisation, clear and accessible user notice, least-privilege access, secure communication and storage, defined retention and deletion policies, incident response, and auditable decision and appeal procedures. These requirements should be considered throughout data collection, model development, deployment, and operation because the system processes sensitive facial and biometric information. The current prototype demonstrates the core verification pipeline, while full production-level governance and compliance controls would require further deployment-specific validation and review.

3.3 Formulation of Solutions

Four solution patterns were considered. A mobile-only model offers low latency and privacy but limited capacity and visibility into injection. A server-only model permits stronger analysis but increases network dependence, data transfer, cost, and central privacy risk. Active challenges add session-specific evidence but create friction and can exclude some users. A risk-based cascade combines a lightweight local path with selective escalation and capture controls.

Table 3.3: Qualitative design decision matrix (5 is favourable). Scores are design judgements and must be validated by field evidence.

Alternative	Security breadth	Mobile latency	Privacy	Accessibility	Cost
Mobile-only passive model	2	5	4	4	5
Server-only strong model	3	2	2	3	2
Mandatory active challenge	4	3	3	2	3
Risk-based layered cascade	5	4	4	4	3

The selected and deployed design is the two-stage system in Figure 3.1. A Google ML Kit liveness challenge (blink, turn, smile) gates capture on-device; a passing clip is scored locally by the MobileNetV3 spoof classifier while the same frame is uploaded in parallel to the server, where a five-way router selects one of five Xception manipulation specialists. The session is accepted only when both verdicts agree the face is real (Table 4.2); either stage failing routes to a flagged/retry state rather than a silent pass, and flagged cases are intended to feed a human-review queue whose interface is not yet built. This is a simpler, evaluated instance of the calibrated cascade formulated in Chapter 1: it uses a hard AND rule rather than confidence bands, and it uses the Xception router as its second-layer evidence rather than G^2V^2former or GD-FAS, which remain candidate alternatives evaluated separately (Chapter 5) but not wired into this call path. Their outputs are not interchangeable with the router's: any future substitution or fusion should be calibrated and tested for complementary error under a common protocol before replacing the deployed second stage.

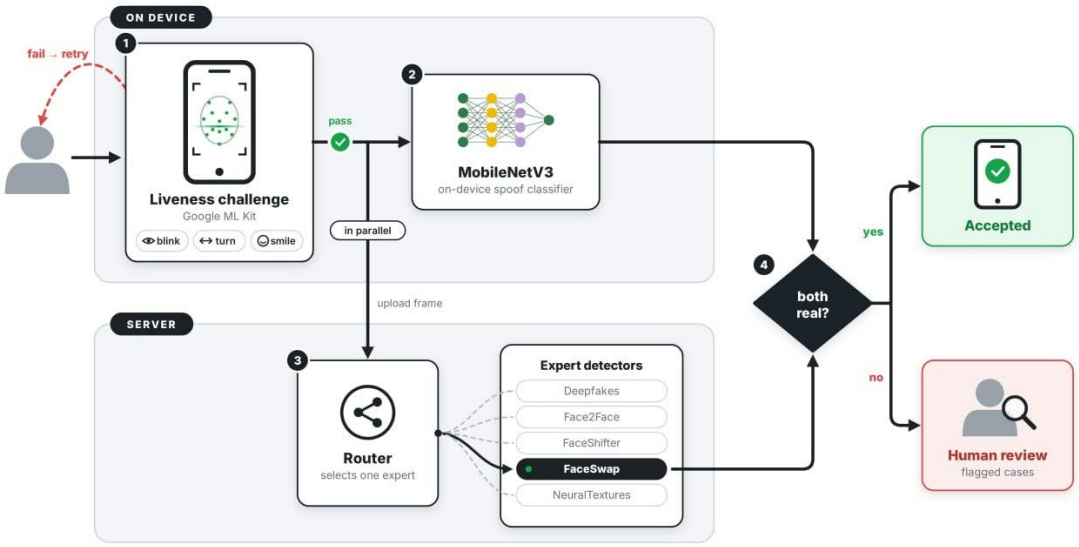


Figure 3.1: Deployed two-stage verification pipeline. Steps 1–2 run on-device; step 3 is the server-side router-and-specialist call; step 4 is the AND decision described in Table 4.2. The dashed retry loop and the human-review box are the only elements not evaluated end to end.

Chapter 4

System Design and Implementation

4.1 Data

Four datasets were used, one per attack family or generalisation question the system is evaluated against. Table 4.1 states each dataset's size, class balance, and split as recorded directly from the pipeline's own run logs, not from the source papers' reported totals, so that every count below is reproducible from the repository.

Table 4.1: Datasets used, with counts taken from the project's own preprocessing and training logs.

Dataset	Attack family	Size and split, as run
FaceForensics++ (c23)	Digital manipulation (5 methods)	600,000 frames from 1,000 source-video pair groups; 900 train / 100 val groups; exactly 90,000 train and 10,000 val frames per class (6 classes: original + 5 fake methods)
LCC-FASD	Physical presentation (print/replay)	992 train videos (133 live, 859 attack) and 342 test videos (42 live, 300 attack); heavily imbalanced toward attack in both splits
Celeb-DF-v2	Digital manipulation (face swap)	890 real videos (590 Celeb-real + 300 YouTube-real) and 5,639 synthesised videos; official test protocol lists 518 videos, of which 100 were evaluated; 6,011 videos remain after removing the official test list, of which 300 were used for fine-tuning (Section 5.1)
Asian-Fakes	Physical presentation, demographic subgroup	4,610 images: 1,996 from an Asian subgroup (998 live / 998 attack) and 2,614 from a non-Asian subgroup (1,307 live / 1,307 attack); each subgroup is exactly class-balanced internally

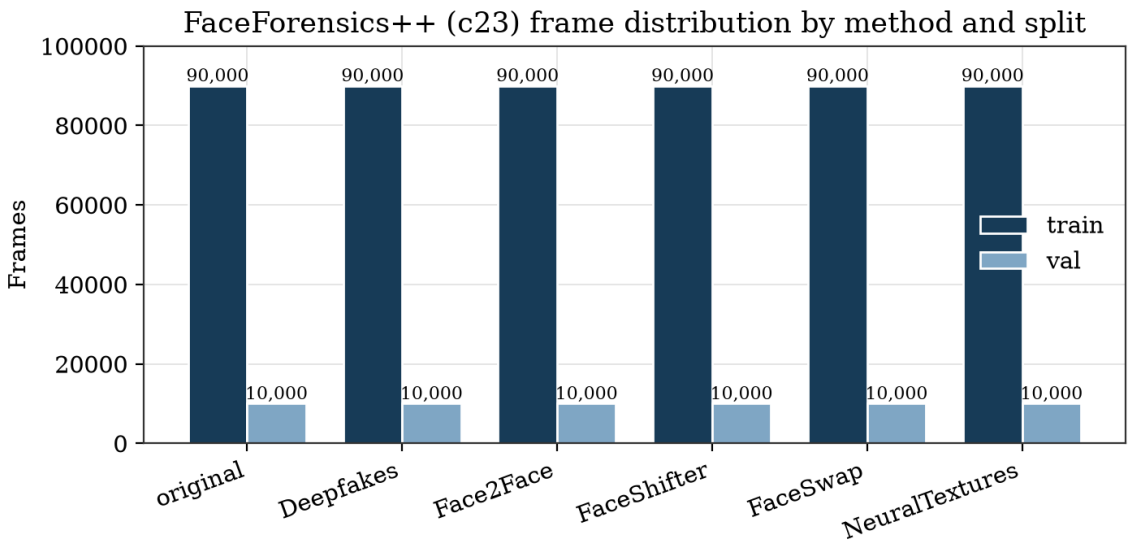


Figure 4.1: FaceForensics++ frame distribution by method and split. The extraction pipeline enforces exact balance across all six classes and both splits, so no class-weighting is needed for this dataset.

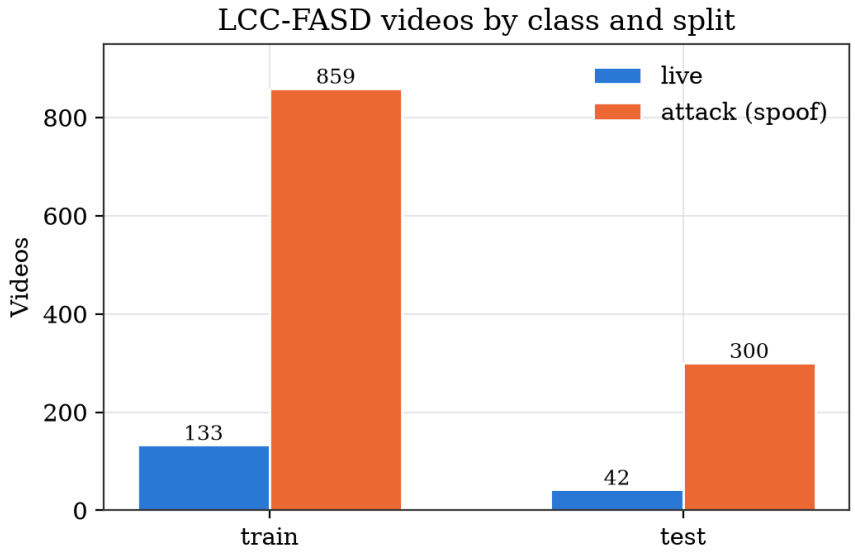


Figure 4.2: LCC-FASD videos by class and split. Attack outnumbers live by roughly 6:1 in train and 7:1 in test, which is why training uses class-weighted loss and why accuracy alone would be misleading (Section 5.1.2).

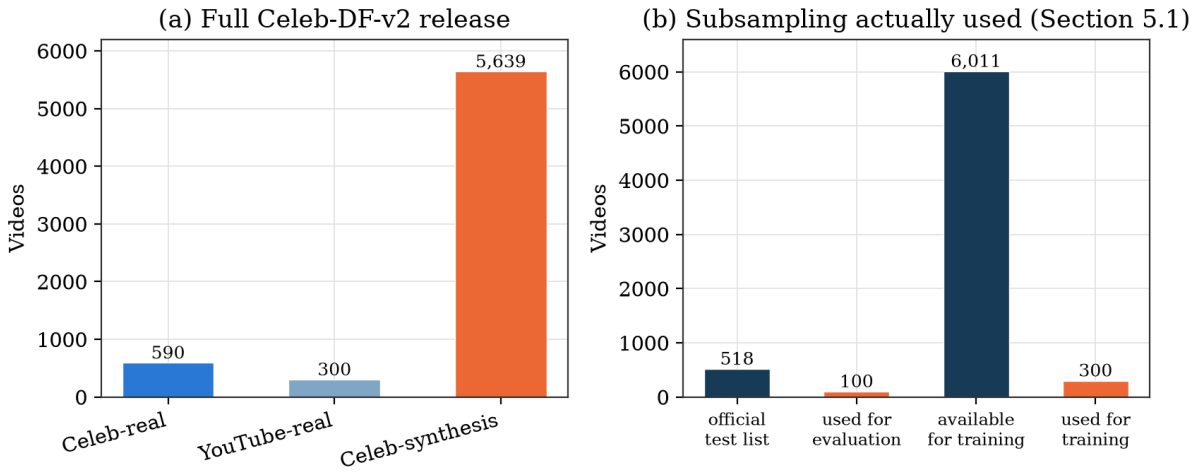


Figure 4.3: Celeb-DF-v2: (a) the full public release is itself imbalanced, roughly 6:1 synthetic to real; (b) the subset actually used for evaluation and fine-tuning in this project (Section 5.1) is a small fraction of what is available—100 of 518 official test videos, 300 of 6,011 available training videos—which is a scale limitation this report states explicitly rather than implying full-benchmark evaluation.

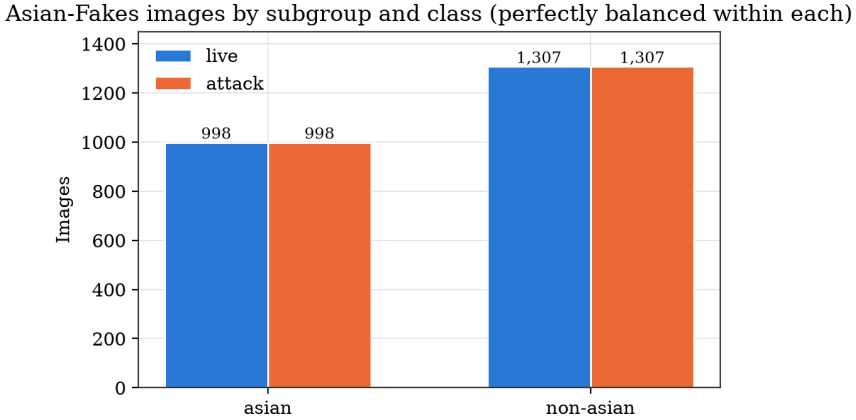


Figure 4.4: Asian-Fakes images by subgroup and class. Each subgroup is internally balanced, which supports a same-scale asian-versus-non-asian comparison, but the two subgroups are unequal in size (1,996 vs 2,614) and the dataset alone cannot establish demographic fairness (Section 6.5.1).

Two limitations follow directly from these distributions rather than from model choice. First, LCC-FASD's and Celeb-DF-v2's attack-heavy imbalance means a model that always predicts "attack" would score misleadingly well on raw accuracy; every result in Chapter 5 therefore reports balanced accuracy, HTER, or AUC rather than accuracy alone. Second, the Celeb-DF-v2 subsampling in Figure 4.3(b) means the cross-dataset results in Table 5.1 describe a 100-video evaluation and a 300-video fine-tuning set, not the dataset's full 518-video protocol or its full 6,011 available training videos; a larger run remains future work (Section 5.3).

4.2 Functional Design and Sub-problem Partitioning

The project was partitioned by evidence and operating layer rather than by one monolithic model:

1. **Capture and preprocessing:** record a short clip, sample frames, locate/crop/normalise faces, and reject unusable capture.
2. **Mobile temporal screening:** compute per-frame MobileNetV3 features, aggregate across the clip, and return a first-layer probability.
3. **Rich temporal analysis:** combine visual tokens and facial-landmark motion in G^2V^2former.
4. **Spatial domain generalisation:** evaluate GD-FAS under source-only and fine-tuned transfer protocols.
5. **Deepfake specialisation:** train a universal Xception baseline and one binary specialist per manipulation method, then learn a meta-router.
6. **Policy and governance:** calibrate scores, select escalation bands, audit decisions, preserve privacy, and provide human fallback.

Temporal information appears at multiple levels, illustrated in Figure 4.6. The lightweight model averages frame embeddings; the transformer represents explicit time/landmark relationships; and the target decision layer can evaluate score stability across the session. The first two have experimental artefacts in the repository; the third is target design.

The as-deployed system runs the mobile and server stages as two independent, parallel evaluators of the same capture rather than a confidence-banded cascade: the on-device MobileNetV3 stage screens for physical presentation attacks while the frame is concurrently uploaded to the server-side router-and-specialist path for a digital-manipulation check. Table 4.2 states the combination rule. It is a conjunction, not an escalation policy: the session is accepted only when both stages independently agree the face is real, and any single "spoof" or "fake" verdict routes the session to a rejected/retry state in the application rather than a silent pass. This is simpler than, and distinct from, the confidence-band cascade with τ_{pass} and τ_{attack} described in Chapter 1: no calibrated risk band decides which cases reach the server, because both stages currently run on every session.

Table 4.2: Combination rule applied to the two independent stage verdicts.

On-device spoof screen	Server deepfake check	Verdict
Real	Real	Accepted
Spoof	Real	Rejected / flagged
Real	Fake	Rejected / flagged
Spoof	Fake	Rejected / flagged

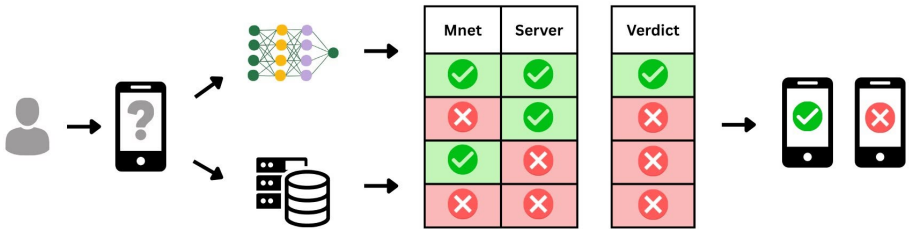


Figure 4.5: The same combination rule as Table 4.2: each stage contributes an independent real/spoof verdict, and only the all-real combination is accepted.

A rejected/flagged outcome currently terminates in an in-app retry prompt. No reviewer console, case queue, or override workflow was found in the repository, so the "human review" stage in Figure 3.1 remains **Proposed**: the application produces the flagged state that a review workflow would consume, but the workflow itself, and the labelled-data feedback loop that would use its outcomes for retraining, are target design rather than implemented artefacts.

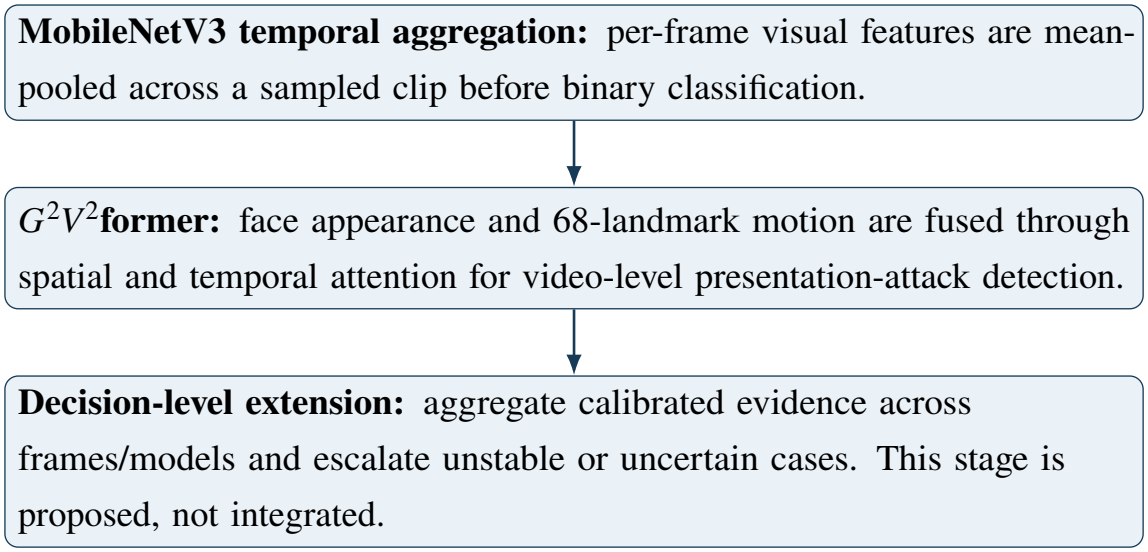


Figure 4.6: Temporal evidence investigated or proposed at three levels.

4.3 Design Refinement and Simulation

The design evolved through successive findings. Early work treated a small mobile model as the primary detector and a server as a stronger fallback. Strong within-domain physical-PAD results motivated explicit temporal research. The G^2V^2former experiments then exposed poor transfer to high-quality digital manipulations, while GD-FAS provided a spatial domain-generalisation comparison. Fine-tuning improved some deepfake results but introduced source-retention and unseen-domain trade-offs. The later Xception study therefore tested whether the heterogeneity of manipulation artefacts was better handled with specialists.

Two refinements follow from this progression. First, "liveness" must include capture provenance: a visually live face can still be injected. Second, the second layer must not be described as one fixed model before common-protocol calibration and ablation show which evidence is complementary. The architecture therefore presents G^2V^2former, GD-FAS, and Xception routing as evaluated candidates, not an already fused product.

4.3.1 Mobile temporal model

For frame x_t, the MobileNetV3 backbone produces feature vector h_t. The clip representation and probability are

$$\bar{h} = \frac{1}{K}\sum_{t=1}^{K} h_t, \qquad p_1 = \sigma(w^\top \bar{h} + b). \tag{4.1}$$

The exported and application contract uses $K = 24$. A training script in the repository uses $K = 10$, so checkpoint provenance and sequence-length consistency must be reconciled before

the exported artefact is associated with a reported training run. Mean pooling is efficient and permutation-invariant, but it cannot represent temporal order.

4.3.2 G^2V^2former and GD-FAS

G^2V^2former uses an eight-frame input in the inspected experiment and 68 facial landmarks per frame. Its explicit spatiotemporal attention makes it a candidate server model, but face/landmark preprocessing dominates a part of latency and introduces an additional failure surface. GD-FAS operates on frames and optimises group-aware/domain-invariant features. The comparison is deliberate: it asks whether temporal dynamics or spatial domain regularisation transfers more reliably across the selected datasets.

4.3.3 Embedding-conditioned specialist routing

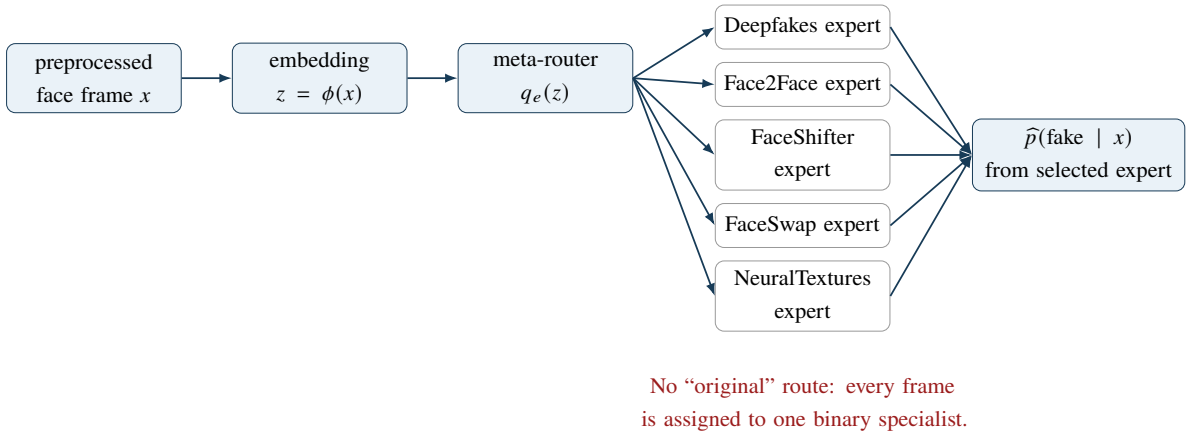


Figure 4.7: Hard specialist-routing formulation as implemented. A meta-model selects one of five binary experts from the frame embedding; it never predicts an original class.

The specialists share an Xception architecture (approximately 20.81 million parameters) but are tuned separately for Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures. Xception's depthwise-separable convolution provides a consistent baseline architecture [8]. The hard routing design in Figure 4.7 computes only the selected expert at decision time if router features are available independently.

The router's training target is a best-expert label rather than a manipulation- method label. Let $s_e(x_i, y_i)$ be expert e's predicted score for training frame x_i against its true class y_i. Every frame is scored by all five already-trained experts, and the label used to train the router is

$$e_i^* = \arg\max_{e \in \mathcal{E}} s_e(x_i, y_i). \tag{4.2}$$

The router is then trained as a five-way classifier over these best-expert labels, with no original/method class and no fallback branch: every frame, real or fake, is assigned to exactly one of the five binary specialists.

4.3.4 Models used, by code artefact

Table 4.3 states, for every model discussed in this report, the exact architecture, its configuration as run, and where its implementation lives in the repository, so that every number in Chapter 5 is traceable to a specific artefact rather than to the reference paper alone.

Table 4.3: Models used in this project, their configuration, and their code location.

Model	Architecture as run	Key configuration	Code artefact
MobileNetV3 (Layer 1)	STDL-FacePAD pattern: shared MobileNetV3 encoder per frame, mean-pooled clip feature, linear head [12]	$K = 24$ frames (export/app); training script records $K = 10$ (Section 5.3); 224×224 RGB, ImageNet normalisation	`spatio-temporal-mnetv3/` (`train_mobilenetv3_celebdfv2_kaggle.py`, exported `.ptl`); `mnetv2-anti-spoof-baseline/` for the earlier single-frame baseline
G^2V^2former	Two-stream (vision + landmark-graph) spatiotemporal transformer, Kronecker temporal attention, MAE-style pretraining [13]	$T = 8$ frames, 68 landmarks/frame, masked-reconstruction pretrain then supervised fine-tune, 20 epochs	`g2v2former/` (`g2v2former-train.ipynb`, `-latency.ipynb`, `-server.ipynb`)
GD-FAS	CLIP ViT-B/16 image/text encoder, Groupwise Scaling Risk Minimisation, Feature Orthogonal Decomposition [14]	Official implementation adapted to LCC-FASD → Celeb-DF-v2 two-experiment protocol	`gd-fas/` (`gd-fas-experiments.ipynb`, `-prepare-offline-bundle.ipynb`)
Universal Xception	Cadene/FF++-style Xception, single binary head	299 × 299, AdamW 10^{-4}, weight decay 10^{-5}, batch 128, up to 15 epochs	`DeepFake-MoE/capstone-xception.ipy`

Model	Architecture as run	Key configuration	Code artefact
Five specialists	Same Xception backbone, one binary head per manipulation method (Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures)	Same training configuration as the universal model, trained independently per method	`DeepFake-MoE/capstone-xception.ipy`
Five-way router (CNN)	Same Xception backbone, five-way linear head over 2,048-dimensional embedding	Same training configuration; five-way cross-entropy over best-expert labels	`DeepFake-MoE/capstone-xception.ipy`
Five-way router (XGBoost)	Gradient-boosted trees on the frozen 2,048-dimensional router embeddings	Ablation of the CNN linear head only; embeddings not retrained	`DeepFake-MoE/xception-xgb.ipynb`

Every model above is either exported to a served artefact (MobileNetV3's `.ptl`, the Xception router and specialists behind the FastAPI/ngrok endpoint measured in Table 5.2) or evaluated in its own notebook with saved result tables (G^2V^2former, GD-FAS). None is described from the reference paper's own reported numbers alone; every metric in Chapter 5 is this project's own run of the code artefact in the right-hand column above.

4.4 Application of Tools

Table 4.4: Main tools and their engineering roles.

Tool	Role	Use in this project
PyTorch / torchvision	Training, transfer learning, evaluation, export	MobileNetV3, G^2V^2former, GD-FAS, and Xception experiments [17].
Flutter / Dart	Cross-platform mobile application	Camera capture, 24-frame preprocessing, PyTorch Lite invocation, and local result display [18].
FastAPI	Research inference service	Separately demonstrated endpoint for the G^2V^2former workflow; not connected to Flutter.
XGBoost	Router-head ablation	Tree-based best-expert prediction from saved embeddings in the five-way routing experiment.
OpenCV / face and landmark tooling	Frame extraction and facial preprocessing	Face crops and landmark sequence construction; failure handling needs hardening.
Jupyter / Colab	Reproducible experiment narrative	Training, evaluation, result tables, and latency pilot; environment/version locking remains incomplete.
Git / GitHub	Source history and collaboration	Version control for source and reports; large datasets/checkpoints require separate governed storage.

4.5 Implementation

The repository contains four main implementation strands. The mobile pipeline includes temporal MobileNetV3 training/export code and a Flutter application. The G^2V^2former strand contains data preparation, adapted training/evaluation, and a FastAPI prototype. The GD-FAS strand contains domain-generalisation evaluation and fine-tuning work. The `DeepFake-MoE` directory contains two Xception notebooks: the main CNN routing experiment and the XGBoost router-head variant. The directory name is retained as a repository label; this report uses

"embedding-conditioned specialist routing" because the current hard, single expert selection is more precise than mixture-of-experts terminology.

The routing dataset was constructed from FaceForensics++ C23 with original and five fake methods. The saved configuration uses 1,000 video groups: 900 training groups and 100 validation groups. Sampling 100 frames per method/video yields 540,000 training frames and 60,000 validation frames. Groups, rather than frames, are separated to reduce video-content leakage. Models take 299×299 images; training includes blur, JPEG compression, horizontal flipping, AdamW with initial learning rate 10^{-4} and weight decay 10^{-5}, batch size 128, up to 15 epochs, and a binary threshold of 0.5.

4.6 Deployment

Two artefacts are deployed and were exercised end to end for this report. The Flutter application captures a clip, creates one 24-frame input, invokes the on-device PyTorch Lite MobileNetV3 model, and applies the fixed 0.5 boundary described in Section 4.1. Concurrently, the frame is sent to a FastAPI service hosted on a Kaggle notebook session and exposed through an ngrok tunnel, which runs the five-way router and the selected Xception specialist and returns a fake probability, the routed class, and per-call latency; Table 5.2 was measured against this live endpoint rather than a static benchmark script. The two verdicts are combined by the client under the rule in Table 4.2. This differs from a production deployment in several respects that remain open: the ngrok endpoint is a development tunnel tied to an interactive notebook session rather than a versioned, authenticated production service; the connection is not currently protected by mutual authentication beyond a static API key; and scores are not calibrated. The G^2V^2former FastAPI service was demonstrated separately and is not part of this call path.

A production deployment would require versioned model contracts; authenticated, encrypted client–server communication; genuine-camera and device-integrity signals; replay protection; strict media retention; rate limiting; monitoring; calibrated policies; rollback; and a reviewer/appeal process. Raw face videos should not be logged by default. Audit records should store the minimum necessary metadata—such as consent/purpose, model version, quality state, calibrated scores, decision reason, and override—under a declared retention schedule.

Chapter 5

Investigation and Evaluation

5.1 Experimental Investigation

5.1.1 Evaluation questions and protocols

The investigation tests three hypotheses: (H1) temporal models can learn useful physical-PAD evidence; (H2) strong source-domain performance will decline under a shift from physical spoofs to high-quality digital manipulation; and (H3) manipulation-specific specialists with a learned router can outperform one universal frame detector on a controlled in-domain split.

The physical/deepfake transfer study uses LCC-FASD, Celeb-DF-v2, and Asian-Fakes. Two saved protocols are compared for each of G^2V^2former and GD-FAS: training on LCC-FASD, and then fine-tuning the resulting model on Celeb-DF-v2. All three datasets are evaluated after each stage. This is a demanding domain shift because physical replay/print cues and digital synthesis cues are not interchangeable. Celeb-DF-v2 itself was constructed to reduce the obvious artefacts of older deepfake data [9].

The Xception study uses FaceForensics++ C23. Validation is group-disjoint from training at the source-video group level. The universal detector sees original and all five fake methods. Each specialist sees original versus one method. Two routers—a CNN head and XGBoost—predict, from the frame embedding, the best-expert label (the specialist expected to score the frame most accurately) and dispatch the frame to that specialist; neither router predicts an original class or has an original-route fallback. This protocol answers an in-domain routing question only; it is not an unseen-generator or production-liveness evaluation.

5.1.2 Metrics

Let bona fide be the negative class and attack/fake the positive class. Accuracy can be inflated by class imbalance, so balanced accuracy and class-conditional rates are reported. For PAD experiments,

$$\text{APCER} = \frac{\text{attacks classified as bona fide}}{\text{all attacks}}, \tag{5.1}$$

$$\text{BPCER} = \frac{\text{bona fide classified as attack}}{\text{all bona fide}}, \tag{5.2}$$

$$\text{HTER} = \frac{\text{APCER} + \text{BPCER}}{2}. \tag{5.3}$$

ROC-AUC measures ranking across thresholds; average precision (AP) is informative under fake-heavy class balance. None of these, by itself, establishes calibrated probability, robustness to injection, or a financially acceptable operating point. ISO/IEC 30107-3 reporting should be followed for a formal PAD evaluation [7].

5.2 Performance Evaluation

5.2.1 Temporal and domain-generalisation results

Table 5.1 gives the results reproduced across the current plan and inspected experiment artefacts. Values are percentages. The source-only G^2V^2former result demonstrates useful LCC-FASD discrimination but transfers poorly to Celeb-DF-v2. GD-FAS is exceptionally strong on its source data yet also falls to near-chance target behaviour. Fine-tuning on Celeb-DF-v2 improves target and Asian-Fakes results for both models, but the target HTER remains high and source performance decreases.

Table 5.1: Common-protocol cross-domain results (%). Lower HTER and higher AUC are better.

		LCC-FASD		Celeb-DF-v2		Asian-Fakes	
Model	Training stage	HTER	AUC	HTER	AUC	HTER	AUC
G^2V^2former	LCC-FASD	22.38	80.71	59.71	37.14	50.13	49.82
G^2V^2former	+ Celeb-DF-v2	33.50	65.14	43.32	57.22	49.13	50.33
GD-FAS	LCC-FASD	**4.55**	**99.69**	52.99	50.49	48.33	52.98
GD-FAS	+ Celeb-DF-v2	7.07	99.16	**41.04**	**64.35**	**44.58**	**57.94**

These results do not show that temporal information is unhelpful. They show that temporal modelling does not remove the need for representative domains and generator diversity. A model may learn the motion of print/replay attacks while missing temporally plausible face swaps. Conversely, GD-FAS's source result does not prove digital manipulation generalisation. The appropriate conclusion is that attack taxonomy, training domains, and evaluation domains must be reported together.

An earlier ROSE-Youtu MobileNetV3 temporal experiment recorded 98.46% accuracy, 99.88% AUC, 1.39% EER, and 1.36% HTER. It is retained as evidence that the lightweight temporal approach can perform strongly on the earlier physical-PAD protocol, but it predates the exported mobile checkpoint and must not be quoted as that checkpoint's performance.

Before the numerical-stability fix described in Section 4, the project's earlier G^{2}V^{2}former-based prototype was evaluated on the 342-video LCC-FASD development split (300 spoof, 42 live), giving 72.8% accuracy, 80.5% ROC-AUC, and 26.6% HTER at an equal-error threshold of 0.239. Figures 5.1 and 5.2 show the resulting confusion matrix and score distribution: 81 of 300 spoof samples were accepted as live, which the earlier report itself judged unacceptable for a security-critical deployment. This result is retained as the honest starting point of the project rather than omitted; the subsequent scatter-mean/clamping fix and 20-epoch fine-tuning run discussed below improved on it, and the separate MobileNetV3 screening stage discussed next improved on it further.

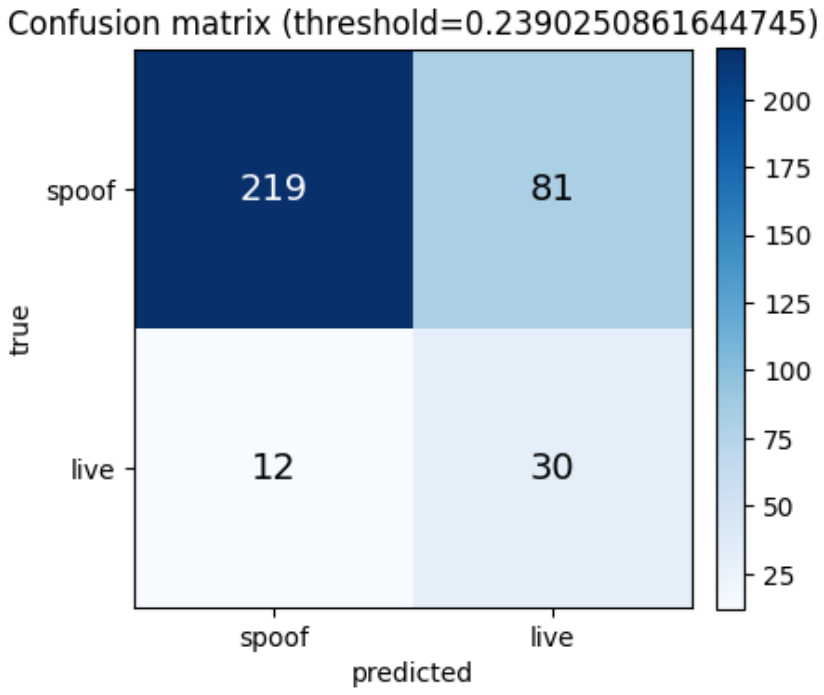


Figure 5.1: Confusion matrix from the preliminary, pre-fix prototype. The 81-sample spoof-accepted-as-live cell is the principal security concern that motivated the extended fine-tuning run below.

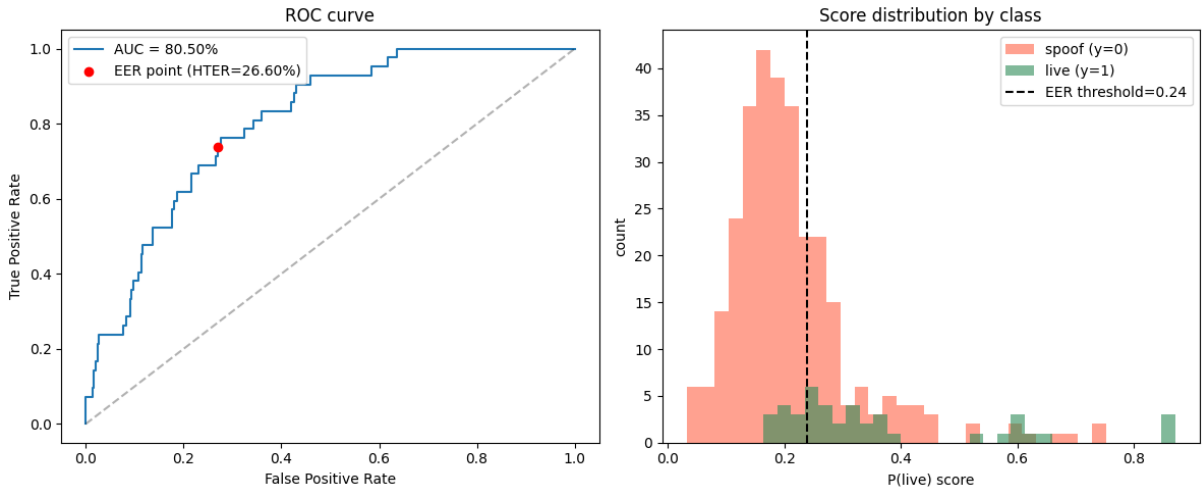


Figure 5.2: ROC curve and score distribution for the same preliminary run. Live samples generally score higher, but substantial class overlap drives the false-acceptance rate above.

The deployed on-device screening stage is a separate model (MobileNetV3 adapted from STDL-FacePAD, Section 2.1) and was evaluated independently on the LCC-FASD test split (7,580 clips), giving 93.8% accuracy and 0.969 ROC-AUC. The two results should not be conflated: the figures immediately above describe the G^{2}V^{2}former-based Layer-2 candidate on 342 development videos, while this figure describes the deployed Layer-1 MobileNetV3 classifier on a larger, separately constructed clip-level test set. This is the figure attributable to the shipped Flutter/PyTorch Lite artefact; the ROSE-Youtu number above remains a separate, earlier-protocol result and the two must not be averaged or substituted for one another. The training-script/ export sequence-length discrepancy described in Section 4 still applies to this checkpoint.

5.2.2 Latency

Table 5.2 reports measured per-frame latency for the FaceForensics++ router-and-specialist path across the environments actually benchmarked. Figures are milliseconds per verification frame, lower is better.

Table 5.2: Measured latency by execution environment, router-plus-specialist path (ms/frame).

Stage / environment	Configuration	Latency (ms)
On-device (mobile)	ML Kit face tracking + MobileNetV3	≈150
Server, CPU	router + one specialist	367
Server, T4 GPU	all seven Xception models (naive)	50.9
Server, T4 GPU	router + one specialist (routed)	14.6
Server, T4 GPU	router only, real-frame fast path	7.3

Two conclusions follow. First, GPU-served routing is materially faster than evaluating every specialist: 14.6 ms against 50.9 ms, a 3.5× reduction, because only the router and the one selected expert run per frame instead of all five. Second, the server path is only worth its network round trip when GPU serving is available—the CPU figure (367 ms) is close to two orders of magnitude slower than the routed GPU path, which is why the server stage should not be assumed interchangeable across hosting tiers without re-measuring. These figures were obtained on the deployed FastAPI/ngrok endpoint described in Section 4 rather than in a research notebook, so they are closer to serving conditions than the G^2V^2former pilot below, though still a single-host measurement rather than a load-tested service-level guarantee.

The inspected G^2V^2former latency pilot used five runs on an NVIDIA T4 and reported a mean of 1,315.45 ms per video end-to-end, of which 134.56 ms was model inference. The small run count and cloud GPU environment make this a diagnostic, not a production latency guarantee. The difference between total and model time also shows that face/landmark preprocessing and data handling deserve optimisation, and it explains why the deployed second stage in Table 5.2 uses the Xception router path rather than G^2V^2former.

The 21.71% figure quoted in the earlier project paper and the 22.38% figure in Table 5.1 are reconciled as two different runs rather than a discrepancy to resolve by rerunning. The earlier paper reports an LCC-FASD-only extended fine-tuning run of 20 epochs after the numerical-stability fix described in Section 4: HTER fell from a 26.6% preliminary-prototype baseline to a best-epoch value of 21.71%, but continued to fluctuate rather than converge (final-epoch HTER 30.98%), which that paper attributes to LCC-FASD's small training partition (992 videos). Figure 5.3 plots this run. Table 5.1 instead reports one checkpoint selected under the later common protocol used to compare G^2V^2former with GD-FAS, so that both models are read at a consistently defined stage rather than each at its own single best epoch; 22.38% is therefore the fairer number for the cross-model comparison in this report, and 21.71% remains the lower bound the architecture reached before common-protocol checkpoint selection was adopted.

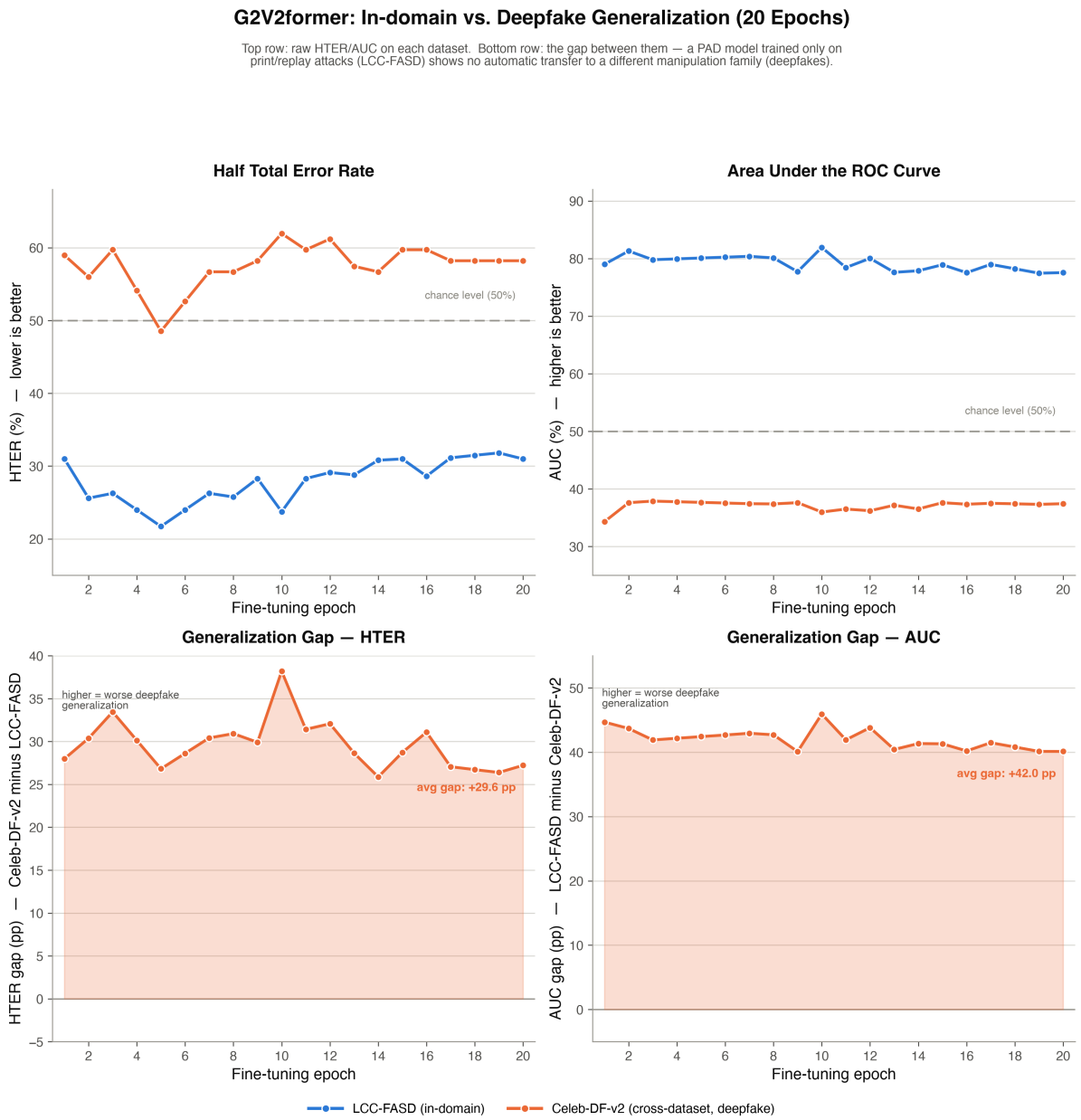


Figure 5.3: HTER and AUC per fine-tuning epoch on the in-domain LCC-FASD split and the held-out Celeb-DF-v2 deepfake benchmark, with the resulting generalisation gap below. The gap averages approximately 30 points of HTER and over 40 points of AUC and does not close with further training. Source: the project's earlier temporal-detection paper.

That earlier zero-shot run offers a more detailed account of the transfer failure than a single-epoch number can: across all 20 epochs, Celeb-DF-v2 HTER stayed in the 55–62% range and its AUC was almost entirely below the 50% chance level, with one exception epoch reaching 48.53% HTER that coincided with the run's best in-domain checkpoint rather than showing a stable trend. The paper's own explanation is that presentation attacks are physical re-capture artefacts—moiré interference, specular reflection, reduced dynamic range, geometric planarity—which are exactly the local photometric and pseudo-3D cues G^2V^2former's visual and landmark streams are built to detect, whereas deepfakes are digital generation-side artefacts (blending-boundary inconsistency,

generator frequency fingerprints, independent-frame temporal flicker) that the architecture, as trained here, has no explicit mechanism to detect. That explanation is retained in this report as the working hypothesis for the domain gap, not as a claim independently verified beyond the cited run.

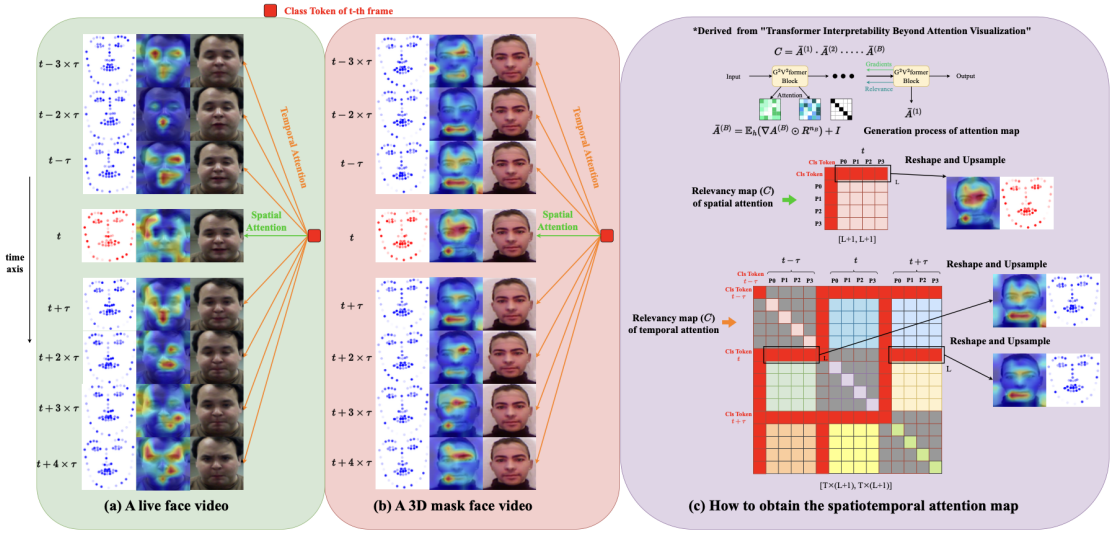


Figure 5.4: Spatial and temporal attention on a live face video (a) versus a 3D mask presentation attack (b): attention concentrates on mask edges and texture, the exact physical re-capture cues discussed above. This is direct evidence of what the model has learned to look for, and why that evidence is the wrong kind for a digital deepfake, which has no mask edge to attend to. Derived from [13] using the attention-relevance method of Chefer et al. (c).

GD-FAS's own mechanism is informative for the same reason. Figure 5.5 shows how its group-wise scaling and Gram-Schmidt orthogonal decomposition align classifier weights and biases *across the presentation-attack source domains it is trained on*: it explicitly targets known domain shift (camera, lighting, subject population) within one attack family, not shift to a different attack family. That is consistent with Table 5.1: GD-FAS's in-domain LCC-FASD result is the strongest of any model in this report, but its zero-shot Celeb-DF-v2 result is close to chance, because the mechanism in Figure 5.5 has nothing to align against a domain it never saw during training. Both architectures, in short, are only as broad as the attack family they were trained on; neither generalises to an attack family its training data and loss function were not built to represent.

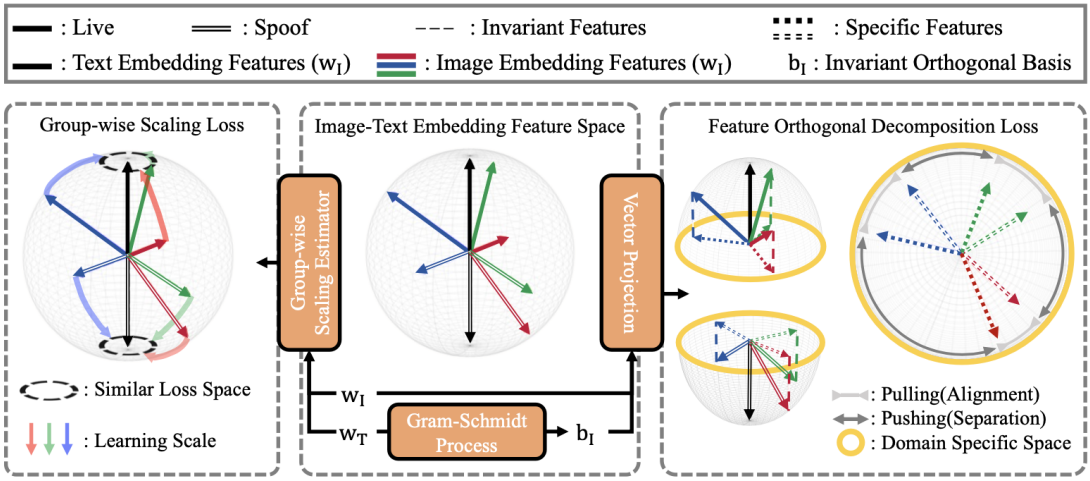


Figure 5.5: How GD-FAS aligns decision boundaries across its presentation-attack source domains: Group-wise Scaling Loss balances per-domain learning scale (left), the Gram-Schmidt process projects live/spoof and domain invariant/specific directions (centre), and Feature Orthogonal Decomposition pulls invariant features together while pushing domain-specific features apart (right). Nothing here references a domain outside the training set. Reproduced from [14].

5.2.3 Universal and routed Xception results

Table 5.3 compares the three saved FaceForensics++ validation results. The routed CNN improves every reported aggregate metric relative to the universal detector, including real-class accuracy. XGBoost also improves balanced accuracy, ROC-AUC, and AP, but underperforms the CNN router.

Table 5.3: Five-way in-domain routing results on the group-disjoint FaceForensics++ validation set (%).

System	Accuracy	Balanced acc.	ROC-AUC	AP	Real acc.	Fake acc.
Universal Xception	81.83	76.44	86.32	96.62	68.37	84.52
CNN-routed specialists	**84.44**	**84.51**	**91.68**	**98.22**	**84.62**	**84.40**
XGBoost-routed specialists	83.49	82.39	88.73	97.73	80.73	84.05

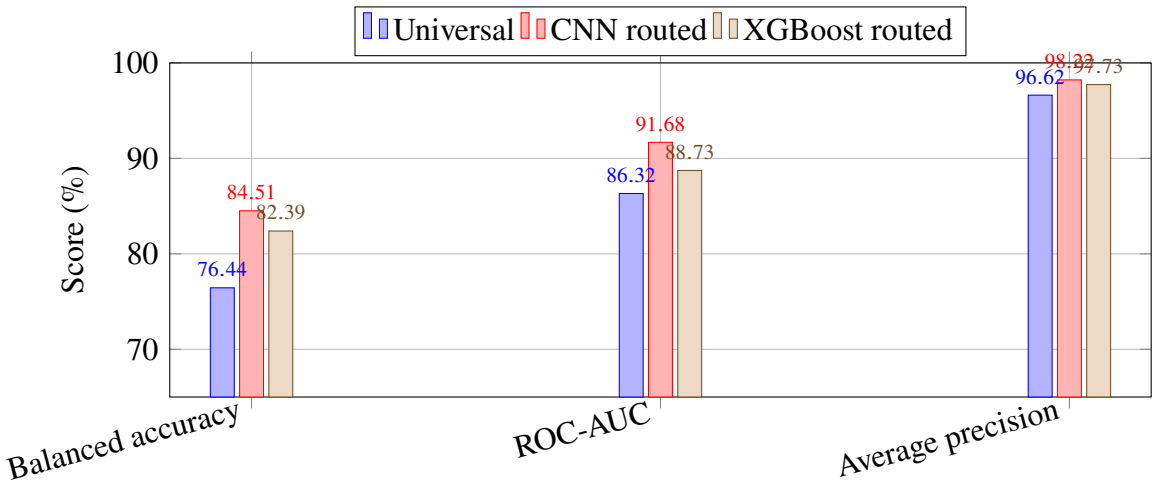


Figure 5.6: Aggregate routing comparison across the universal detector and the five-way best-expert routed CNN and XGBoost systems.

Per-method balanced accuracy in Table 5.4 explains why the idea is promising. Every specialist outperforms the universal detector for its method, and routing recovers part, but not all, of that oracle-like advantage. Router mistakes therefore remain a measurable bottleneck. NeuralTextures is the hardest of the five classes for the specialist and routed systems.

Table 5.4: Per-manipulation balanced accuracy of specialist, universal, and current routed CNN systems (%).

Manipulation	Matched specialist	Universal	Routed CNN
Deepfakes	94.86	79.81	88.62
Face2Face	92.36	78.47	85.20
FaceShifter	93.24	75.70	87.64
FaceSwap	90.37	75.58	84.98
NeuralTextures	80.28	72.66	76.11

The result supports a narrow conclusion: under the saved group-disjoint, in-domain FF++ protocol, method-conditioned specialist selection improves a universal Xception baseline. It does not demonstrate robustness to unseen generators, external datasets, recapture, injection, video-level aggregation, or deployment demographics. It also incurs an expert-training and storage cost that must be compared with stronger universal or multi-task baselines.

5.2.4 Assessment against objectives

Table 5.5: Objective-level evaluation status.

Objective	Status	Evidence-based assessment
Mobile temporal prototype	Partial success	Integrated 24-frame local path; checkpoint/training provenance and device profiling remain.
Temporal deepfake transfer	Investigated	G^2V^2former results expose severe domain shift; not yet operationally adequate.
Domain-generalised comparison	Investigated	GD-FAS strong in-domain, limited cross-domain; fine-tuning improves target but remains weak.
Specialist routing	Achieved	Five-way best-expert routing outperforms the universal baseline on the in-domain, group-disjoint FF++ validation split.
Layered end-to-end system	Not completed	Mobile and API artefacts exist separately; fusion, escalation, review, and IAD are target design.
Fairness, privacy, and security validation	Not completed	Risks and requirements are documented; demographic, field, red-team, and governance evaluation remain.

5.3 Deviations and Design Revisions

The main deviations are substantive and should remain visible in the final report:

1. The planned three-tier system was narrowed to an integrated mobile first stage and separately tested server candidates; no reviewer interface was completed.
2. The temporal MobileNet training code uses 10-frame clips in one path, whereas export/inference/application code uses 24; the final artefact cannot inherit earlier metrics without provenance evidence.
3. G^2V^2former checkpoint metrics differ between the latest PDF and saved artefacts and require one reproducible final evaluation.
4. Strong physical-PAD performance did not transfer to digital deepfakes, motivating GD-FAS comparison and manipulation-specific Xception work.
5. Capture-channel attacks became an explicit system concern. Media analysis alone is no longer represented as sufficient protection from injection.

These revisions improve the report's validity: they separate demonstrated evidence from architecture intent and convert unresolved contradictions into testable follow-up work.

Chapter 6

Project Management, Teamwork, and Impact

6.1 Project Management and Finance

6.1.1 Planning and Risk Management

The work progressed from problem framing and a three-tier target architecture to mobile temporal modelling, richer temporal analysis, domain-generalisation comparison, integration, and finally specialist routing. Figure 6.1 shows this sequence in relative phases.

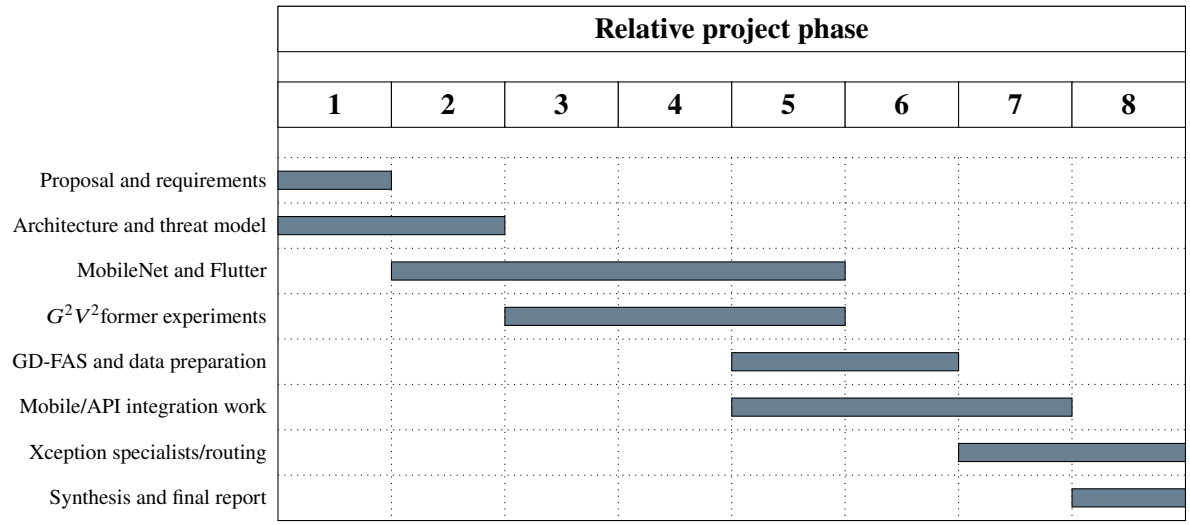


Figure 6.1: Project progression. Phase indices show order and overlap.

Table 6.1: Principal project risks and mitigations.

Risk	Consequence	Mitigation and remaining action
Frame/identity leakage	Inflated validation performance	Split by source-video group; audit every dataset at identity and derivation level.

Risk	Consequence	Mitigation and remaining action
Domain or generator shift	Failure on new attacks despite high source accuracy	Cross-dataset evaluation, diverse training, unseen-generator hold-out, drift monitoring, conservative escalation.
Class imbalance	Misleading accuracy and unsafe threshold	Report balanced accuracy and class rates; calibrate on deployment prevalence/cost.
Router failure	Correct expert is bypassed	Report oracle/specialist gaps, top-k or soft-routing ablations, and a safe universal fallback.
Capture injection	Classifier receives convincing forged media	Device/sensor attestation, protected channel, replay defence, IAD and red-team testing.
Biometric disclosure	Irreversible privacy harm	Minimise collection, encrypt, separate identifiers, restrict access, delete by policy, incident response.
False rejection	Exclusion and repeated exposure	Accessible recapture, bounded retry, alternative verification, timely appeal and review.
Training instability or missing checkpoint	Irreproducible claims	Freeze environment/configuration/checksum and regenerate result tables from artefacts.
Model obsolescence	New manipulations evade detection	Versioned monitoring, red-team sets, retraining triggers, rollback, and maintained ownership.
Scope inflation	Prototype presented as product	Maintain implemented, experimental, and proposed labels in architecture and claims.

6.1.2 Budgeting and Resource Identification

An earlier weekly report planned a USD 260–540 GPU budget, made in anticipation of dataset and cost support from Pathao, the company that originally proposed this project. That support did not materialise: Pathao indicated it would provide datasets and cover costs, but communication

stopped before either was delivered, and the planned budget was never drawn on. The team instead completed all development, experimentation, and storage using publicly available datasets, personally owned devices, and Kaggle's free-tier GPU notebooks, incurring no cash cost. The one exception is the live demonstration: the deployed system was hosted for one day on a paid DigitalOcean virtual machine so it could be shown running end to end rather than only in notebooks.

Table 6.2: Resource and budget record.

Resource	Recorded cost	Evidence/status
GPU compute	USD 0	Kaggle's free-tier GPU notebooks were used for all training, experimentation, and (via an ngrok tunnel) the demo inference endpoint; the earlier USD 260–540 estimate assumed paid compute that was never needed.
Local development devices	USD 0	Personally owned laptops/phones; no incremental project cost beyond existing ownership.
Storage and network	USD 0	Public dataset downloads and artefact storage stayed within Kaggle's and personal devices' free capacity; no paid storage or bandwidth service was used.
Software	USD 0 licence cost for main tools	Primarily open-source research and development stack; licence obligations still apply.
Demo deployment	USD 0.43	One-day DigitalOcean Basic Droplet (2 GiB RAM, 1 vCPU, USD 0.01786/hour × 24 hours) used only to host the live demonstration; not a sustained or production deployment.

Total recorded cash expenditure for the project is therefore USD 0.43, entirely attributable to the one-day demonstration deployment; every other resource used public data, free-tier compute, or personally owned equipment.

6.1.3 Economic Analysis and Financial Prospecting

The cascade is economically attractive only if local screening reduces server and review cost without increasing fraud or abandonment beyond acceptable limits. For N sessions, a transparent

expected-cost model is

$$C = N(C_{\text{edge}} + r_s C_{\text{server}} + r_h C_{\text{review}}) + N_{\text{FA}} L_{\text{fraud}} + N_{\text{FR}} L_{\text{friction}}, \tag{6.1}$$

where r_s and r_h are escalation and review rates. The loss terms must be estimated from partner evidence rather than borrowed percentages. Calibration curves can then show how changing a threshold moves cost between compute, review, false acceptance, and false rejection.

No financial return is claimed in this project because production prevalence, fraud loss, abandonment, reviewer cost, and integration cost were not supplied. The useful output is the model above and the measurements needed to populate it.

6.1.4 Sustainability in Business and Commercialisation

A responsible path is: reproducible research validation; licence/privacy/security review; partner-specific data collection with consent; red-team and independent PAD/IAD evaluation; monitored shadow deployment; a limited pilot with appeal; and only then a staged rollout. Long-term ownership must cover model monitoring, incident response, reviewer training, data retention, accessibility, compliance, and retirement of obsolete models. A specialist bank increases maintenance cost; its gain must persist against a stronger universal model before commercial use.

6.2 Individual and Teamwork

6.2.1 Participation and Contribution

Contribution of each team member is listed here.

Table 6.3: Contribution record to be verified by the team.

Candidate	Student ID	Verified concrete contributions and delivery evidence
Nafis Nahian	2105007	MobileNetV3 and deepfake-detection experiments: implemented and evaluated the lightweight MobileNetV3 temporal face-anti-spoofing model, including preprocessing, training, and performance analysis; contributed to the deepfake-detection experiments through Xception-based model training, dataset preparation, evaluation, and analysis of universal and manipulation-specific detectors; contributed to experimental debugging, result interpretation, integration, and report preparation.

Candidate	Student ID	Verified concrete contributions and delivery evidence
Abrar Zahin Raihan	2105009	Mobile application and system integration: built the Flutter capture flow (short front-camera clip, 24-frame preprocessing); integrated the exported MobileNetV3 PyTorch Lite model for on-device inference and result display; added the ML Kit active liveness challenge (blink/turn/smile) that gates capture; implemented the parallel client–server call to the FastAPI router/specialist endpoint and the client-side AND verification rule with the flagged/retry flow; tested the deployed pipeline end to end and measured on-device latency ($\approx$150 ms/frame).
Aritra Debnath	2105010	MobileNetV2, active-liveness, and deepfake-detection experiments: implemented and evaluated the MobileNetV2-based lightweight face-anti-spoofing model, including preprocessing, training, and performance analysis; worked on the Google ML Kit active-liveness component using facial actions such as blink, head turn, and smile; contributed to the deepfake detection pipeline through Xception-based training and evaluation, manipulation-specific specialist experiments, result analysis, integration testing, debugging, and report preparation.
Tamzeed Mahfuz	2105012	GD-FAS and deepfake-detection experiments: implemented and evaluated the GD-FAS domain-generalisation pipeline, including source-domain experiments, cross-domain evaluation, fine-tuning, and analysis of transfer between physical presentation attacks and digital manipulations; contributed to Xception-based deepfake-detection experiments, dataset preprocessing, manipulation-specific detector evaluation, debugging, result analysis, literature review, and report preparation.
Ruwad Naswan	2105051	G2V2Former temporal face-anti-spoofing experiments: implemented and evaluated the G2V2Former pipeline, including video and facial-landmark preprocessing, model setup, training, inference, and evaluation; investigated spatial and temporal attention for motion-based liveness detection and analysed within-domain and cross-domain behaviour; contributed to inference/API experiments, latency evaluation, debugging, result interpretation, literature review, and report preparation.

6.2.2 Collaboration and Conflict Resolution

The project required close collaboration because the individual model experiments eventually had to support a common liveness-verification system. Team members regularly discussed dataset preparation, evaluation protocols, model checkpoints, metric definitions, and the interfaces between the mobile and server components. Experimental results and implementation issues were shared among the team so that problems in one component could be reviewed by members working on the other components.

Technical disagreements were resolved primarily through the available experimental artefacts and additional testing rather than by choosing one interpretation arbitrarily. When strong performance on physical presentation-attack data did not transfer well to deepfake data, the team revised the experimental direction instead of assuming that a single model would generalise to both attack types. This led to further investigation using GD-FAS and manipulation-specific Xception detectors. These decisions affected the final system design but helped ensure that reported conclusions were consistent with the experimental evidence.

6.2.3 Leadership and Direction

Leadership within the team was distributed according to the main technical components of the project rather than being assigned to a single student throughout the entire development process. Nafis Nahian took primary responsibility for the MobileNetV3 experiments and also contributed to the deepfake-detection work. Abrar Zahin Raihan led the Flutter application, user-interface, and system-integration work. Aritra Debnath worked primarily on MobileNetV2, the ML Kit active-liveness component, and deepfake detection. Tamzeed Mahfuz led the GD-FAS experiments and contributed to the deepfake detection studies, while Ruwad Naswan took primary responsibility for the G2V2Former experiments. Members reviewed results across these workstreams and worked together during integration, debugging, evaluation, and preparation of the final report.

The technical direction also evolved in response to experimental findings. Poor cross-domain transfer motivated the team to investigate domain generalisation and manipulation-specific detection rather than relying on the initial models alone. The supervisor provided overall research guidance and proposed the specialist-routing idea, while its implementation, evaluation, integration, and analysis were carried out collaboratively by the team. As the project progressed, the team prioritised reproducibility and integration of completed components, while features that could not be sufficiently validated were clearly separated from the implemented system and retained as future work.

6.3 Impact on Society

Reliable remote verification can widen access, reduce impersonation, and allow a customer to complete a sensitive process without travel. The same system can also deny service, intensify surveillance, or expose biometric data. Impact is therefore conditional on who can use the capture flow, whose faces and devices were represented during evaluation, what happens after rejection, and whether a customer can challenge a decision.

False acceptance may harm the impersonated person and the provider. False rejection may disproportionately burden people with older cameras, unstable connectivity, darker or lighter skin under poor illumination, facial differences, religious attire, disability, or difficulty performing an active challenge. Repeated retry can itself increase biometric exposure and frustration. Socially responsible deployment requires an equivalent alternative verification route, clear notice, bounded retry, timely human appeal, accessible instructions, and regular publication of aggregate error and override behaviour where safe.

6.4 Environmental Impact and Life Cycle Sustainability

GPU training, repeated hyperparameter search, dataset replication, network transfer, server escalation, and retained video all consume energy and storage. The cascade may reduce serving energy by resolving clear cases locally, but mobile battery use and the embodied impact of devices still matter. A five-expert bank also multiplies training and storage unless sharing or distillation is used.

The environmental plan is to record GPU type and hours; reuse frozen embeddings for router ablations; stop unpromising runs early under a declared rule; cache preprocessing safely; select the smallest model meeting validated requirements; limit media retention and duplication; batch server work where latency allows; and measure energy/latency per accepted, escalated, and reviewed transaction. Lifecycle planning must include model replacement and secure deletion, not only initial training.

6.5 Ethics

Ethical analysis is a design constraint throughout the pipeline. The main risks, controls, and missing evidence are shown in Table 6.4.

Table 6.4: Ethical risk–control–evidence matrix.

Risk	Required control	Evidence still required
Biometric privacy	Purpose limitation, minimisation, encryption, separated identifiers, least privilege, deletion, breach process	Data inventory, retention schedule, access audit, deletion test, incident exercise
Inadequate consent	Plain-language notice of purpose, processing, retention, sharing, alternatives, and appeal	Comprehension/usability study and approved notice
Demographic disparity	Representative evaluation, subgroup error/calibration, intersectional testing, mitigation and stop criteria	Consented demographic protocol and sufficiently powered results
Device/network disparity	Quality-aware capture, low-bandwidth behaviour, supported-device testing, alternative path	Device matrix and field trials
False rejection	Bounded retry, accessible recapture, reason categories, alternative verification and human appeal	FRR by condition/group; appeal time, overturn and abandonment rates
False acceptance	Capture integrity, PAD/IAD, red-team tests, conservative high- risk policy and monitoring	Attack-presentation and injection test campaign
Selective escalation bias	Audit who is escalated/reviewed and outcome gaps; calibrate by policy without discriminatory shortcuts	Escalation, review, and override rates by relevant subgroup and device condition
Reviewer automation bias	Show uncertainty/evidence limits, train reviewers, blind irrelevant attributes, sample QA and second review	Human-factors study and inter-reviewer consistency

Risk	Required control	Evidence still required
Dual-use disclosure	Publish enough for reproducibility while withholding operational secrets that materially ease bypass; responsible vulnerability route	Threat-informed release and disclosure plan
Dataset/IP misuse	Licence/provenance audit, minimum redistribution, attribution, and removal process	Dataset-by-dataset permission record

6.5.1 Equity and Inclusivity

Aggregate AUC can conceal unequal false-accept and false-reject rates. Evaluation should pre-register relevant groups and capture conditions, avoid using sensitive attributes as unexplained proxies, report uncertainty, and investigate intersections where sample size permits. Asian-Fakes provides one additional domain but is not, by itself, proof of demographic fairness. A face dataset's name or geographic origin does not guarantee a balanced or consented fairness sample.

The user interface should not require speech, hearing, a narrow range of facial motion, expensive hardware, or sustained high bandwidth without an equivalent alternative. Escalation and review are also fairness surfaces: a group sent to review more often experiences delay and greater human scrutiny even if the final acceptance rate appears equal.

6.5.2 Accountability and Personal Responsibility

Every decision should be attributable to a model version, configuration, threshold policy, quality state, and responsible owner. Model developers are responsible for reporting limitations; product owners for appropriate use; security teams for capture/channel controls; reviewers for reasoned decisions; and management for resourcing appeal and incident response. A human reviewer must not become an accountability sink: override rules, evidence access, second review, quality sampling, and appeal outcomes require governance.

The candidates accept responsibility for distinguishing completed work from proposal, retaining contradictory results until reconciled, and correcting the routing formulation. High performance on one benchmark must never be advertised as comprehensive fraud prevention.

6.5.3 Proper Use of Intellectual Property

Table 6.5: Third-party asset and attribution inventory.

Asset	Use	Required action
FaceForensics++, Celeb-DF-v2, LCC-FASD, Asian-Fakes, ROSE-Youtu	Research data	Verify access terms, citation, redistribution, consent/provenance, and derivative frame policy separately.
G^2V^2former and GD-FAS	Adapted research methods/code	Cite papers and repositories; preserve licence notices; document modifications and weights.
MobileNetV3 and Xception	Model architectures/pretrained components	Cite original work; record library/weight licence and version.
PyTorch, Flutter, FastAPI, XGBoost and supporting libraries	Implementation stack	Maintain dependency bill of materials and comply with their licences.
BUET logo and report template	Institutional report material	Use only for the course report and preserve required formatting/attribution.
Generative-AI assistance	Drafting/editing support	Disclose per course policy; students verify all content and do not assign authorship to the tool.

No dataset, code, or figure should be redistributed merely because it is publicly accessible. The final repository release should include only artefacts allowed by their licences and access agreements.

6.5.4 Professionalism and Ethical Codes

The ACM and IEEE codes emphasise avoiding harm, honesty about system limitations, respect for privacy, fairness, competence, and responsibility [15, 16]. Applied here, professionalism means resisting pressure to turn benchmark numbers into deployment guarantees; documenting negative and contradictory results; protecting captured faces; obtaining expertise outside machine learning; and suspending or rolling back a system when monitored harm exceeds agreed thresholds. Security details should be disclosed responsibly, and affected users should receive meaningful explanation and recourse.

Chapter 7

Conclusion

7.1 Conclusion and Future Work

This capstone investigated liveness verification as a layered engineering problem, not as a single binary-classification benchmark. The implemented work spans a lightweight temporal MobileNetV3 mobile path, explicit temporal modelling with G^2V^2former, domain-generalised spatial analysis with GD-FAS, public-data preparation and transfer experiments, a separate server prototype, and later manipulation-specific Xception routing. Together these components reveal both the promise and the current limits of robust remote face verification.

The objectives were met unevenly but informatively. A 24-frame local inference flow is integrated in Flutter, although its checkpoint provenance and device performance need a final audit. G^2V^2former and GD-FAS were evaluated across physical-PAD and deepfake domains. Their results show that excellent source-domain performance can collapse under a change in attack family, and that target-domain fine-tuning improves some results without solving unseen-domain generalisation. The later FF++ experiment shows that matched specialists and learned, five-way best-expert routing can improve an in-domain universal Xception baseline: the CNN-routed artefact raises balanced accuracy from 76.44% to 84.51% and ROC-AUC from 86.32% to 91.68%. That finding is promising but narrow, because it has not been tested against unseen generators, injection, or deployment populations.

The strongest project conclusion is therefore methodological. Physical presentation attacks, digital manipulation, and capture-channel injection require different but coordinated evidence. Temporal reasoning can contribute, domain generalisation must be measured rather than assumed, and manipulation specialists may help when their routing targets and evaluation are leakage-safe. None of these replaces genuine-sensor assurance, protected transport, calibrated policy, privacy protection, subgroup evaluation, or accountable recourse.

The next work should proceed in the following order:

1. freeze an experiment manifest and reconcile the MobileNetV3 sequence length/checkpoint and G^2V^2former metric discrepancies;
2. confirm the best-expert router's training labels were generated out-of-fold, with a declared

tie rule and a policy for frames on which all experts fail, and re-report the routing result if they were not;

3. compare the best-expert router against oracle routing, top-k/soft routing, and a stronger universal or multi-task baseline;
4. evaluate all deepfake systems at video level on unseen generators, additional compression/recapture, and independent external datasets;
5. calibrate scores and produce reliability, threshold-cost, and escalation- coverage curves rather than adopting a universal 0.5 threshold;
6. connect the mobile and server components through an authenticated, versioned contract and implement failure-safe offline/network behaviour;
7. add genuine-camera confidence, replay protection, device integrity, and injection red-team scenarios rather than relying on media analysis alone;
8. conduct representative device, demographic, accessibility, privacy, and reviewer studies with explicit stop criteria and appeal measurement; and
9. evaluate latency, memory, bandwidth, energy, serving/review cost, licence compliance, monitoring, rollback, and secure deletion in a governed pilot.

A human-review path should be implemented only with trained reviewers, bounded evidence access, reason codes, second-review rules, workload testing, and user appeal. It should not be described as a generic cure for model uncertainty.

7.2 Continuous Engagement and Self-directed Learning

The project required continuous learning across research and engineering boundaries. The team moved from an initial mobile/server concept to video transformers and landmark motion, then to domain-generalisation methods, mobile model export, API prototyping, Xception specialists, and routing ablations. Reading primary papers was necessary but insufficient: implementations had to be adapted, dataset structures reconciled, numerical issues debugged, checkpoints interpreted, and results compared under common metrics.

Negative transfer results were particularly valuable. They changed the project from a search for one high score into an investigation of attack taxonomy, dataset shift, and evidence boundaries. The injection threat similarly expanded the design beyond classifier accuracy to capture provenance and channel security. The routing discrepancy reinforced a further lesson: a mathematically attractive claim is valid only when labels, data splits, code paths, and evaluation actually implement it.

Professional development should continue through reproducible experiment packaging, biometric security and privacy study, mobile profiling, secure deployment, calibration, human-factors evaluation, and responsible disclosure. Future engagement with a financial partner should begin with jointly defined threats, lawful and consented data governance, operational error costs, and pilot stop conditions. These practices are as important to trustworthy liveness verification as further model optimisation.

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